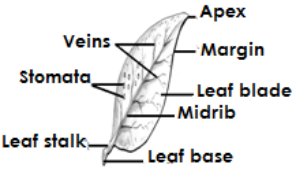
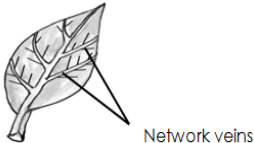
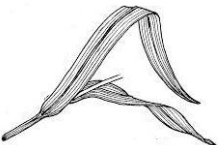


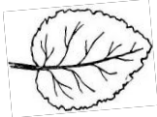


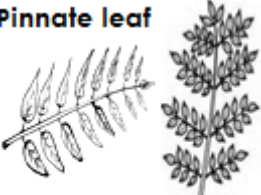
PRIMARY FOUR SCIENCE SCHEME OF WORK FOR TERM I, 2025

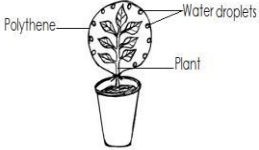
W K	P D	TOPIC	TOPIC	CONTENT	SUBJ. COMP'NCES	LANG. COMP'NCES	METHODS	ACTIVITIES	IND. OF L.S.V	INST. MAT.	REF.	RE M.
1	1	Plant life	Floweri ng plants	Flowering plants - Flowering plants are plants that bear flowers. Example of flowering plants - Maize Plants - Bean Plants - Banana Plants - Tea plants - Coffee Plant Characteristics of flowering plants - Flowering plants bear flowers. - Most flowering plants reproduce by means of seeds. Structure of a flowering plant - Stem - The root system - The terminal bud - Axillary bud - Node and internode	The learner; 1. Defines flowering plants 2. Gives examples of flowering plants. 3. States the characteristics of flowering plants. The learner; 1. Draws the structure of a flowering plant. 2. names the parts of a flowering plant 3. Identifies the systems of a flowering plant. 4. Draws and names parts of a flowering plant	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to flowering plants e.g. Bear, reproduce, and flower. 2. Reads, internalize and writes texts and questions related to flowering plants.	Guided discovery Discussion observatio n	Defining flowering plants Giving examples of flowering plants Drawing the structure of a flowering plant. Naming parts of a flowering plant.	Appreciati- on. Care Awareness Fluency Concern	Floweri ng plants	Mk. Int. sci. pbk 4 Int. sci. syllab us bk 4	
1	2	Plant life	Structu re of a leaf	Structure of a leaf	The learner; 1. Draws the structure of a leaf. 2. Names the	The learner; 1. Pronounces, spells, reads writes and	Guided discovery Discussion	Drawing the structure of a leaf	Appreciati- on. Care	Differen t types of leaves	Mk. Int. sci. pbk 4	

				 Functions of parts of a leaf 1. Leaf stalk - Holds the leaf on the branch 2. Midrib - Transports manufactured food and water. 3. Leaf blade - Has stomata used for breathing/ gaseous exchange	parts of a leaf. 3. Mentions the functions of each part of a leaf.	demonstrates meaning of words related to flowering plants e.g. Bear, reproduce, and flower. 2. Reads, internalize and writes texts and questions related to flowering plants.	Observation	Naming parts of a leaf. Mentioning the functions of parts of a leaf.	Awareness Fluency Concern	brought in the class.	Int. sci. syllabus bk 4	
1	3	Plant life	Leaf venation	Leaf venation - Leaf venation is the arrangement of veins in a leaf. Types of leaf venation 1. Network leaf venation 2. Parallel leaf venation. NET WORK LEAF VENATION ⇒ The leaf has veins made like a net. Illustration	The learner; 1. Defines the term leaf venation. 2. Identifies the types of leaf venation. 3. Give examples of plants the network and parallel leaf venation.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to leaf venation e.g. network and parallel 2. Reads, internalize and writes texts and questions related to leaf venation.	Guided discovery Discussion Observation	Defining leaf venation Identifying types of leaf venation. Giving examples	Appreciation. Care Awareness Fluency Concern	Different types of leaves brought in the class	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

1				 Network veins ⇒ Network venation is found in leaves of dicotyledonous plants. Example of plants with network leaf venation Bean, Hibiscus, Flower, Peas Parallel leaf venation ⇒ The veins in a leaf run from the leaf stalk to the apex of the leaf in a parallel form.  ⇒ Parallel venation is found in leaves of monocotyledonous plants like; Maize, Millet, Sugarcane, Rice, Grass and wheat plants.	The learner; 1. Defines the term leaf venation. 2. Identifies the types of leaf venation. 3. Give examples of plants the network and parallel leaf venation.		Guided discovery Discussion observation	of leaf venation Defining leaf venation			Mk. Int. sci. pbk 4	Int. sci. syllabus bk 4	
1	4	Plant	Types	TYPES OF LEAVES	The learner	The learner;	Guided	Identifying	Appreciati-	Differen	Mk.		

		life	of leaves	1. Simple leaves 2. Compound leaves SIMPLE LEAVES A simple leaf is a leaf with one leaf blade and one leaf stalk. Examples of simple leaves - Simple serrated - Simple lobed leaf - Simple entire - Simple divided entire - Simple palmate Simple serrated  Simple divided entire   etc.	1. Identifies the types of leaves. 2. Describes simple leaves. 3. Mentions examples of simple leaves. 4. Draws the different structures of simple leaves	1. Pronounces, spells, reads writes and demonstrates meaning of words related to leaf venation e.g. network and parallel 2. Reads, internalize and writes texts and questions related to simple leaves	discovery Discussion observation	the types of leaves. Describing simple leaves. Mentioning examples of simple leaves. Drawing the structure of simple leaves	on. Care Awareness Fluency Concern	t types of leaves brought in the class	Int. sci. pbk 4 Int. sci. syllabus bk 4	
1	5	Plant life	Types of leaves	Compound leaves Types of leaves Compound leaves ⇒ A compound leaf	The learner 1. Describes simple leaves. 2. Mentions	The learner; 1. Pronounces, spells, reads writes and demonstrates	Guided discovery	Describing comp. leaves.	Appreciation.	Different types of comp.	Mk. Int. sci. pbk 4	

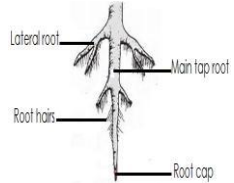
				is a leaf with many leaflets on one main leaf stalk. ⇒ The leaflets are divided at original leaf stalk. ⇒ Each leaflet has its own small stalk called rachis Examples of compound leaves 1. Compound pinnate leaf 2. Compound bipinnate leaf 3. Compound trifoliate leaf 4. Compound digitate leaf Illustration  Pinnate leaf Bi-pinnate	examples of simple leaves. 3. Draws the different structures of compound leaves	meaning of words related to leaf venation e.g. pinnate, bi-pinnate and trifoliate 2. Reads, internalize and writes texts and questions related to compound leaves	Discussion observation	Mentioning examples of comp. leaves. Drawing the structure of simple leaves	Awareness Fluency Concern	leaves brought in the class	Int. sci. syllabus bk 4	
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				Trifoliate  Digitate 								
	6	Plant life	Transpiration the rate of transpiration.	TRANSPIRATION - Transpiration is the process by which plants lose water in form of water vapour to the atmosphere. Importance of transpiration a) To a plant 1. It cools the plant. 2. It enables plants to absorb more water from the soil. b) To the environment ⇒ It helps in the formation of rain fall.  Factors which affect the rate of	The learner; 1. Defines transpiration. 2. Gives the importance of transpiration to; i) Plants ii) Animals.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to transpiration 2. Reads, internalize and writes texts and questions related to transpiration.	Guided discovery Discussion observation	Defining transpiration. Giving the importance of transpiration to plants and the environment.	Appreciation. Care Awareness Fluency Concern	The local environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

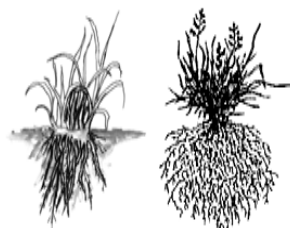
				transpiration 1. Temperature 2. Light intensity 3. Humidity 4. Wind 5. Surface area of a leaf							
1	7	Plant life	Ways plants reduce the rate of transpiration.	Ways in which plants reduce the rate of transpiration ⇒ By shedding their leaves in the dry season to reduce the rate of transpiration. ⇒ Some plants develop a layer of wax on their leaves. ⇒ Some plants reduce the size of their leaves to thorny structures. ⇒ Some plants develop thick leaves to store water for the dry season. ⇒ Some plants modify their leaves into thorns e.g. cactus plants. Dangers of	The learner 1. Mentions ways in which plants reduce the rate of transpiration. 2. State the dangers of transpiration.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to transpiration 2. Reads, internalize and writes texts and questions related to transpiration	Guided discovery Discussion observation	Defining transpiration. Giving the importance of transpiration to plants and the environment.	Appreciation. Care Awareness Fluency Concern	The local environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4

				transpiration ⇒ Excess transpiration leads to wilting of plants.								
2	1	Plant life		PHOTOSYNTHESIS ⇒ Photosynthesis is the process by which plants make their own food. Requirements for photosynthesis 1. Water 2. Carbon dioxide 3. Sunlight 4. Chlorophyll The raw materials for photosynthesis 1. water 2. carbon dioxide. Conditions necessary for Photosynthesis 1. Chlorophyll 2. Sunlight 1. Importance of photosynthesis to people 1. Due to photosynthesis, people get food. 2. People get oxygen for respiration due to	The learner; 1. Defines the term photosynthesis. 2. Mentions the requirements for photosynthesis. 3. Describes the conditions, raw materials, product and by product for photosynthesis.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to photosynthesis e.g. chlorophyll, sunlight & water. 2. Reads, internalize and writes texts and questions related to Photosynthesis	Guided discovery Discussion Observation Brain storming Think pair and share	Defining photosynthesis. Mentioning the requirements for photosynthesis. Describing the conditions, raw product and by products of photosynthesis.	Appreciation. Care Awareness Fluency Concern	The local environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				photosynthesis.								
2	2	Plant life	Uses of leaves	Functions (uses) of leaves to plants ⇒ Leaves make food for the plant. ⇒ Leaves help a plant to carry out transpiration. ⇒ Some leaves store food and water for the plant e.g. onions ⇒ Some leaves are used for plant propagation e.g. bryophyllum Uses of leaves to people ⇒ Some leaves are eaten as food e.g. Amaranthus. ⇒ Some leaves are used as herbal medicine. ⇒ Some leaves are used to thatch houses. ⇒ Plant leaves are used for research and study purposes.	The learner 1. Gives the uses of leaves to people. 2. Mentions the importance of leaves to plants.	The learner 1. Pronounces, spells, reads writes and demonstrates meaning of words related to uses of leaves. 2. Reads, internalize and writes texts and questions related to uses of leave	Guided discovery Discussion Observation Brain storming	Giving the uses of leaves to people. Mentioning the uses of leaves to plants.	Appreciation Care Awareness Fluency Concern	The local environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
2	3	Plant	roots	Root system	The learner;	The learner	Guided	Naming	Appreciation	Grass	Mk.	

		life		ROOTS ⇒ A true root system develops from the radicle of the embryo. Types of root system 1. Tap root system 2. Fibrous root system a) Tap root system  1. Root hairs: absorb water and mineral salts from the soil. 2. Root cap: Protects the growing tip of a root. Examples of plants with tap root system Mangoes, beans, b) Fibrous root system ⇒ This is the type of root system where there are many roots growing randomly from	1. Names the systems of roots. 2. Draws a well labelled structure of a tap root and fibrous root systems.	1. Pronounces, spells, reads writes and demonstrates meaning of words related to root systems. 2. Reads, internalize and writes texts and questions related to root systems.	discovery Discussion Observation Brain storming	types of root systems. Drawing a well labelled structure of root systems.	on. Care Awareness Fluency Concern	carrots	Int. sci. pbk 4 Int. sci. syllabus bk 4	
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the radicle of a seed
Illustration

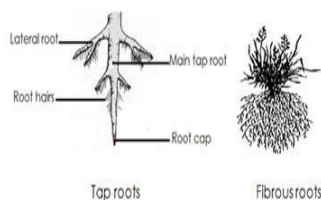


TYPES OF ROOTS

1. Primary roots
 2. Secondary roots
- a) Primary roots**
⇒ These are roots that develop from the radicle of a germinating seed.

Examples of primary roots

1. Tap roots
2. Fibrous roots



b) Secondary roots (adventitious roots)

The learner;

1. Identifies the types of roots.
2. Describes primary and secondary roots.
3. Mentions examples of primary and secondary roots.
4. Draws the structure of a prop root.

The learner

1. Pronounces, spells, reads writes and demonstrates meaning of words related to types of roots
2. Reads, internalize and writes texts and questions related to types of roots.

Guided discovery

Discussion

Observation

Brain storming

Identifying the types of roots.

Describing primary and secondary roots.

Mentioning examples of primary and secondary roots.

Appreciation.

Care

Awareness

Fluency

Concern

Cassava tubers, sweet potato tubers, carrots

Mk. Int. sci. pbk 4

Int. sci. syllabus bk 4

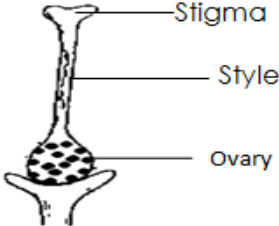
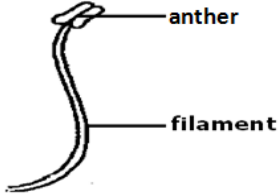
				⇒ These are roots that develop from any other part of the plant other than the radicle. Examples of adventitious roots 1. Prop root 2. Stilt roots 3. Buttress roots 4. Storage roots 5. Clasping roots 6. Breathing roots								
2	5	Plant life	Uses of roots	USES OF ROOTS TO PLANTS ⇒ Roots fix the plant firmly in to the soil. ⇒ They absorb water from the soil. ⇒ Some roots store food for the plant. Uses of roots to man - Some roots act as food. - Some roots are used as herbal medicine. - Some roots	The learner; 1. states the uses of roots the plant, man and the soil/ environment The learner; 1. states the uses of roots the plant, man and the soil/ environment	The learner 1. Pronounces, spells, reads writes and demonstrates meaning of words related to uses of roots 2. Reads, internalize and writes texts and questions related to uses of roots.	Guided discovery Discussion Observation Brain storming	Stating the uses of roots to plants, people and soil.	Appreciation. Care Awareness Fluency Concern	Cassava tubers, sweet potato tubers, carrots.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				provide fire wood to people when dry. - Some roots are sold for money. Uses of roots to man - Roots hold the soil particles together hence controlling soil erosion.								
2	6	Plant life	Stems	Stems Types Of Stems 1. Upright or erect stems 2. Underground stems 3. Weak stems. 1. Upright Or Erect stems. ▪ They grow straight in space e.g. woody plants, 2. Underground or storage stems ⇒ These are stems	The learner; 1. Names the types of stems 2. Gives the examples of the different types of stems. 3. Describes the categories of underground stems.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to stems. 2. Reads, internalize and writes texts and questions related to stems	Guided discovery Discussion Observation Brain	Naming the types of stems. Giving examples of the different types of stems. Describing the different	Appreciation. Care Awareness Fluency Concern	Different types of stems availed in class.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				which grow underground and store food e.g. Stem tubers, Bulbs & Rhizomes Characteristics of underground STEMS ⇒ They have scale leaves. Categories of underground stems a) Stem tubers e.g. white yams and Irish potatoes. b) Corms e.g. cocoyam, crocus and gladiolus. c) Bulbs e.g. onions, garlic and Spider lily d) Rhizomes e.g. ginger, turmeric, cannalilly and grass such as coach grass.			storming	categories of underground stems.				
2	7	Plant life	Stems	Weak stems ⇒ These are weak stems which	The learner; 1. Describes weak stems.	The learner; 1. Pronounces, spells, reads writes	Guided discovery	Describing plants with weak	Appreciation.	Different types of	Mk. Int. sci.	

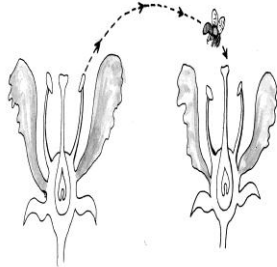
				cannot support themselves upright. ⇒ To get enough sunlight.	2. Gives examples of weak stems. 3. Mentions ways plants with weak stems use to climb others.	and demonstrates meaning of words related to stems e.g. erect, weak stems and underground stems. 2. Reads, internalize and writes texts and questions related to stems.	Discussion	Giving examples of plants with weak stems.	Care	stems availed in class.	pbk 4	
				Groups of weak stems 1. Climbing stems e.g. passion fruits, 2. Creeping stems			Observation	Mentioning ways used by weak stems to climb others.	Awareness			
				How weak stems climb others 1. By using tendrils ⇒ The lateral bud of a plant or the leaf tip develops into a tendril.			Brain storming		Fluency			
				Examples of plants which use tendrils Passion fruits, gourds, cowpeas, pumpkins, cucumber					Concern			
				2. By using hooks ⇒ Some plants have downward pointing thorns.								
				3. By twining or clasping. ⇒ Plants clasp their stems around a support.								

3	1	Plant life	Functions of stems	FUNCTIONS OF A STEM TO A PLANT 1. They hold the leaves on a plant. 2. They hold the flowers for proper pollination. USES OF STEMS TO ANIMALS 1. Most stems are used for timber and firewood. 2. Some stems are used as herbal medicines. 3. Stems are used as food for animals. 4. People use stems as firewood.	The learner; 1. gives the uses of stems to ; a) plants b) people c) animals	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to stems e.g. erect, weak stems and underground stems. 2. Reads, internalize and writes texts and questions related to functions of stems	Guided discovery Discussion Observation Brain storming	Giving the uses of stems to plants, people and animals..	Appreciation. Care Awareness Fluency Concern	Weak stems collected from the environment.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
3	2	Plant life	flowers	FLOWERS ⇒ A flower is a reproductive part of a flowering plant. The external	The learner; 1. Defines a flower. 2. Draws the internal parts of a flower. 3. Name parts of	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to parts of a flower.	Guided discovery Discussion	Defining a flower. Drawing and labeling	Appreciation. Care	Flowers from the environment.	Mk. Int. sci. pbk 4	

				structure of a flower - Sepals - petals The internal structure of a flower - pistil - stamen a) Pistil  b). The Stamen ⇒ The stamen is the male part of a flower. Illustration 	a flower.	2. Reads, internalize and writes texts and questions related to parts of a flower.	Observation Brain storming	the parts of a flower.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
3	3	Plant life	Functions of parts of a	FUNCTIONS OF EACH PART i) Petals ⇒ Petals are	The learner; 1. States the function of each part of a flower.	The learner; 1. Pronounces, spells, reads writes and demonstrates	Guided discovery	Sating the function of each part of a flower.	Appreciation. Care	The environment..	Mk. Int. sci. pbk 4	

			flower	brightly coloured to attract pollinators Pistil or Carpel ⇒ The pistil is the female part of a flower ⇒ It is made up of three main parts; namely Ovary, Stigma and Style USES OF FLOWERS TO PEOPLE ⇒ Flowers are used to get insecticides. ⇒ They are used for decoration on various functions. ⇒ They are used to get dye. ⇒ Flowers are used to make perfumes.		meaning of words related to parts of a flower. 2. Reads, internalize and writes texts and questions related to parts of a flower.	Discussion Observation Brain storming		Awareness Fluency Concern		Int. sci. syllabus bk 4	
3	4	Plant life	Pollination	POLLINATION ⇒ Pollination is the	The learner 1. Defines	The learner; 1. Pronounces,	Guided discovery	Defining pollination.	Appreciation.	Flowers from	Mk. Int.	

				transfer of pollen grains from the anthers to the stigma. TYPES OF POLLINATION 1. Self-pollination 2. Cross pollination Self-pollination ⇒ This is the transfer of pollen grains from the anthers of a flower to the stigma of the same flower or another flower on the same plant. Illustration  HOW THE FLOWER IS ADAPTED (SUITED) FOR SELF POLLINATION ⇒ The anthers and stigma mature at the same time e.g. the conifers. ⇒ The flower remains	pollination. 2. Names and describes the types of pollination. 3. States the adaptation of some flowers to self pollination.	spells, reads writes and demonstrates meaning of words related to pollination. 2. Reads, internalize and writes texts and questions related to pollination.	Discussion Observation Brain storming	Naming the types of pollination. Describing self pollination. Stating the adaptation of some flowers to self pollination.	Care Awareness Fluency Concern	the environ- ment. A chart showing an illustration of self pollination.	sci. pbk 4 Int. sci. syllabus bk 4	
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				closed until self-pollination has taken place e.g. the conifers								
3	5	Plant life	Cross pollination	Cross pollination ⇒ This is the transfer of pollen grains from the anthers of a flower to the stigma of another flower on another plant of the same species. Illustration  HOW THE FLOWER IS ADOPTED (SUITED) FOR CROSS POLLINATION ⇒ The male and female flowers occur on the same plant e.g. in maize ⇒ The male and	The learner; 1. Describes cross pollination. 2. States the adaptation of some flowers to cross pollination. 3. Identifies the agents of pollination.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to cross pollination. 2. Reads, internalize and writes texts and questions related to cross pollination.	Guided discovery Discussion Observation Brain storming	Describing cross pollination. Stating the adaptation of some flowers to cross pollination. Identifying the agents of pollination.	Appreciation. Care Awareness Fluency Concern	Flowers from the environment. A chart showing cross pollination	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

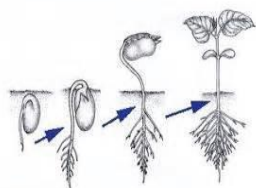
				female flowers occur on separate plants e.g. in papaws. AGENTS OF POLLINATION ⇒ Agents of pollination are things responsible for the transfer of pollen grains to the stigma. Agents of pollination 1. Insects 2. wind 3. water 4. Animals e.g. bats 5. Birds								
3	6	Plant life	CCCs of insect/ wind pollinated flowers	CHARACTERISTICS OF INSECT POLLINATED FLOWERS ⇒ They are scented flowers. ⇒ They produce less sticky pollen. ⇒ They are have brightly coloured petals. ⇒ CHARACTERISTICS OF WIND POLLINATED	The learner; 1. Mentions the CCCs of wind and insect pollinated flowers. 2. States the importance of pollination.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to pollination. 2. Reads, internalize and writes texts and questions related to pollination.	Guided discovery Discussion Observation	Mentioning the CCCs of wind and insect pollinated flowers. Stating the importance of pollination.	Appreciation. Care Awareness Fluency Concern	Flowers from the environment. A chart showing cross pollination	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				FLOWERS. ⇒ The petals have dull colours. ⇒ The flowers do not produce nectar. ⇒ They produce a lot of pollen grains. IMPORTANCE OF POLLINATION ⇒ Pollination allows fertilization to take place in farmers' crops. ⇒ Pollination allows high yield in farmers' harvest.			Brain storming	Identifying the agents of pollination.				
3	7	Plant life	SEEDS	SEEDS A seed is a fertilized mature ovule. Classification of seeds a) Monocotyledonous seeds b) Dicotyledonous seeds Monocotyledonous	The learner; 1. Defines a seed. 2. Mentions types of seeds. 3. Describes monocotyledonous seeds. 4. States the CCCs of monocots with relevant	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to seeds like dicot and monocots. 2. Reads, internalize and writes texts and	Guided discovery Discussion	Defining a seed. Mentioning the types of seeds. Describing monocotyl	Appreciation. Care Awareness Fluency	Different grains such as maize, millet, sorghum.	Mk. Int. sci. pbk 4 Int. sci.	

				⇒ Monocotyledonous seeds are seeds that have one cotyledon. Examples of monocotyledonous seeds 1. maize 2. millet 3. sorghum 4. barley 5. rice A structure of maize grain	examples.	questions related to seeds	Observation Brain storming	edonous seeds. Stating the CCCs of Monocots with relevant examples.	Concern		syllabus bk 4	
4	1	Plant life	Dicotyledonous	Dicotyledonous seeds ⇒ These are seeds that have two cotyledons. Examples include 1. beans 2. peas 3. soya 4. Ground nuts A structure of a bean seed Functions of each part	The learner; 1. Describes dicotyledonous seeds. 2. Gives examples of dicotyledonous seeds. 3. States the CCC of dicotyledonous seeds.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to dicotyledonous seeds. 2. Reads, internalize and writes texts and questions related to dicotyledonous seeds.	Guided discovery Discussion Observation	Describing dicotyledonous seeds. Giving examples of dicotyledonous seeds. Stating the CCCs of	Appreciation. Care Awareness Fluency Concern	Bean seeds, soya or ground nut seeds.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				1. Plumule – develops into shoot system 2. Radicle – develops into root system 3. Testa– protects the internal parts from damage. 4. Cotyledon- It stores food for the seed. 5. Endosperm stores food for the embryo			Brain storming	dicotyledonous seeds				
4	2	Plant life	Seed Germination	Seed Germination ⇒ Germination is the development of a seed embryo into a young plant. ⇒ A seedling is a young plant Condition for germination Water - Softens the testa for the radicle to come out. Warmth - for respiration Oxygen - provides the right temperature	The learner; 1. Defines the term germination. 2. Mentions the conditions for seed germination. 3. Describes the importance of each condition necessary for seed germination.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to seed germination 2. Reads, internalize and writes texts and questions related to Seed germination.	Guided discovery Discussion Observation Brain	Defining the term germination. Mentioning the conditions for seed germination. Describing the	Appreciation. Care Awareness Fluency Concern	Germinating seeds. A chart showing seed germination.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				for the germinating seed. Steps under gone by a seed during germination ⇒ The seed absorbs water through the micropyle and swells. ⇒ Testa softens swells and opens for the radicle to pass out. ⇒ The radicle comes out of the seed to form the root system. ⇒ The Plumule comes out to grow into the shoot system.			storming	importance of each condition of seed germination.				
4	3	Plant life	Types of germination	Types of germination - Epigeal germination - Hypogeal germination Epigeal germination - This is the type in which the cotyledon comes out of the ground. A bean seed	The learner; 1. The learner identifies the types of germination. 2. Describes each type of germination. 3. Illustrates to show the two types of germination. 4. Mentions	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to seed germination 2. Reads, internalize and writes texts and questions related to Seed germination.	Guided discovery Discussion Observati	Identifying the types of seed germination. Describing epigeal germination.	Appreciation. Care Awareness Fluency	A chart showing seed germination.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk	



Examples of seeds that undergo epigeal germination

Beans Ground nuts

Hypogeal germination

- This is type of germinations in which a cotyledon remains under ground.

Illustration

Examples of seeds that undergo hypogeal germination

Maize
Rice

examples of seeds that undergo each type of germination.

on

Brain storming

Illustrating the types of seed germination.

Mentioning examples of seeds that undergo each type of seed germination.

Concern

4

4	4	Plant life	Seed viability and seed dormancy	Seed viability and seed dormancy - Seed viability is the ability of a seed to germinate given the necessary conditions. Characteristics of	The learner; 1. Defines seed viability. 2. States the CCCs of a viable seed. 3. Gives the meaning of	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words seed viability and seed dormancy germination	Guided discovery Discussion	Defining seed viability. Stating the CCCs of a viable	Appreciation. Care Awareness	A chart showing an experiment on seed viability seed vi	Mk. Int. sci. pbk 4	
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				seed viability ⇒ They should be mature. ⇒ They should be free from pest damage. Factors that make a seed fail to germinate ⇒ When the seed has holes. ⇒ When the embryo is not mature. Experiment on seed viability	seed dormancy. 4. States the factors that may make a seed fail to germinate under normal conditions.	2. Reads, internalize and writes texts and questions related to seed viability and seed dormancy	Observation Brain storming	seed. Giving the meaning of seed dormancy. Stating the factors that may make a seed fail to germinate.	Fluency Concern		Int. sci. syllabus bk 4	
4	5	Plant life	Uses of plants	Uses of plants to people ⇒ Some plants are sources of food. ⇒ Some plants are used as herbal medicine. ⇒ Coniferous plants are used for decorating live fences. ⇒ Plants provide firewood and charcoal. ⇒ Some plants	The learner; 1. state the uses of plants to people	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related seed viability and seed dormancy germination 2. Reads, internalize and writes texts and questions related to seed viability and seed	Guided discovery Discussion Observation	Stating the uses of plants to people..	Appreciation. Care Awareness Fluency Concern	Environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				provide hard wood for timbers. ⇒ Plants provide oxygen to animals. Uses of plants to animals ⇒ Some plants provide food to animals. ⇒ Plants provide shelter to some animals. ⇒ Plants provide animals with oxygen for respiration. Importance of plants to the environment ⇒ Plants help in the formation of rainfall. ⇒ Plants help to purify air by absorbing carbon dioxide.		dormancy	Brain storming					
4	6	Growing crops	crops	⇒A crop is a plant that is grown and cared for a particular purpose. Groups of crops (categories of	The learner; 1. Names the examples of common crops. 2. Identifies the	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to crops.	Guided discovery	Naming examples of crops. Identifying	Appreciation. Care	Environment	Mk. Int. sci. pbk 4	

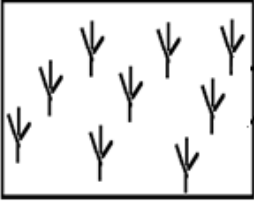
				crops) 1. Annual crops 2. perennial crops a) Annual crops: These are crops which mature and harvested within one year. Examples of annual crops Sun flower sorghum Beans Peas maize Ground nuts Perennial crops: These are crops that are harvested year after year. Examples of perennial crops Coffee crop tea crop banana crop.	groups of crops. 3. Mention examples of crops under each group.	2. Reads, internalize and writes texts and questions related to crops.	Discussion Observation Brain storming	the groups of crops. Mentioning examples of crops under each group.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
4	7	Crop growing	Garden tools	Garden tools Examples of garden tools. - Hoe - Spade - Axe - Rake	The learner; 1. mentions the common tools Draws their structures and states their uses.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to garden tools	Guided discovery Discussion	Mentioning the common tools. Drawing	Appreciation. Care Awareness	Illustrations of different garden tools.	Mk. Int. sci. pbk 4	

				- Wheel barrow - Watering can - Slasher Their structure and uses.		2. Reads, internalize and writes texts and questions related to garden tools.	Observation Brain storming	the structure of different garden tools. Stating the uses of different garden tools.	Fluency Concern	Where possible , real garden tools brought to class.	Int. sci. syllabus bk 4	
5	1	Crop growing	More garden tools	More Garden tools Examples of garden tools. - Forked hoe - Watering can - Trowel - Garden fork - Pick axe Their structure and uses.	The learner; 1. mentions the common tools 2. Draws their structures and states their uses.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to garden tools 2. Reads, internalize and writes texts and questions related to garden tools.	Guided discovery Discussion Observation Brain storming	Mentioning the common tools. Drawing the structure of different garden tools. Stating the uses of different garden tools.	Appreciation. Care Awareness Fluency Concern	Illustrations of different garden tools. Where possible , real garden tools brought to class.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
5	2	Crop growing	More garden tools	More Garden tools Examples of garden	The learner; 1. mentions the	The learner; 1. Pronounces,	Guided discovery	Mentioning the	Appreciation.	Illustrations of	Mk. Int.	

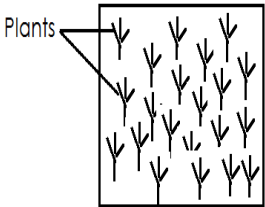
		g	n tools	tools. - More garden tools - Secateur - Pruning saw - Knives - Sickle Their structure and uses.	common tools 2. Draws their structures and states their uses.	spells, reads writes and demonstrates meaning of words related to garden tools 2. Reads, internalize and writes texts and questions related to garden tools.	Discussion Observati on Brain storming	common tools. Drawing the structure of different garden tools. Stating the uses of different garden tools.	Care Awareness Fluency Concern	differen t garden tools. Where possible , real garden tools brought to class.	sci. pbk 4 Int. sci. syllab us bk 4	
5	3	Crop growin g	Caring for garde n tools	Caring for garden tools ⇒ By cleaning them after use. ⇒ By keeping them in a cool dry place. ⇒ By using the tools for their rightful purposes. Rusting ⇒ Rusting is a process by which iron combines with oxygen and moisture to form	The learner; 1. Mentions ways of caring for garden tools. 2. Describes the conditions for rusting. 3. Identifies the ways of preventing rusting.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to care for garden tools 2. Reads, internalize and writes texts and questions related to care for garden tools.	Guided discovery Discussion Observati on	Mentioning the common tools. Drawing the structure of different garden tools. Stating the uses of	Appreciati-on. Care Awareness Fluency Concern	Lubricat ing oil and grease brought to class.	Mk. Int. sci. pbk 4 Int. sci. syllab us bk 4	

				a brown coloured substance called rust. Conditions that favour rusting to take place 1. Oxygen 2. Moisture Dangers of rusting on metals 1. Rusting makes tools weak and worn out. 2. Rusting makes some tools blunts thus leading to wearing due to increased friction. Control of rusting ⇒ By painting metallic garden tools. ⇒ By keeping tools in cool dry places. ⇒ By oiling or greasing some tools.			Brain storming	different garden tools.				
5	4	Crop growing	Crop Growing Practic	Crop Growing Practices ⇒Crop growing practices are	The learner, 1. Mentions the crop growing practices.	The learner; 1. Pronounces, spells, reads writes and demonstrates	Guided discovery	Mentioning the crop growing practices.	Appreciation. Care	The environment	Mk. Int. sci. pbk 4	

			es	activities involved in the production of food e.g. 1. Land preparation 2. Selecting seeds for planting 3. Planting or sowing 4. Transplanting Land preparation - This is the first stage in a food path usually done in the dry season. Activities involved during land preparation ⇒ Cutting down trees using an axe or panga. ⇒ Controlled bush burning. Ploughing land ⇒ It is done using a tractor, ox-plough, hoes, and forked hoe. ⇒ Ploughing the land is done to make the soil loose and soft	2. Describes land preparation. 3. States the different activities done during land preparation.	meaning of words related to crop growing practices. 2. Reads, internalize and writes texts and questions related to crop growing practices.	Discussion Observation Brain storming	Describing land preparation Stating the different activities done during land preparation	Awareness Fluency Concern		Int. sci. syllabus bk 4	
5	5	Crop growin		Planting ⇒ This refers to	The learner; 1. Describe the	The learner; 1. Pronounces,	Guided discovery	Describing the term	Appreciation.	The school	Mk. Int.	

		g		putting seeds in holes and covering with soil. Example of planting materials - Seeds, suckers, stem cuttings, bulbs Qualities of good planting materials ⇒ They should be mature and healthy. ⇒ The seeds should not be broken. ⇒ They should be of desired characteristics. Methods of planting 1. Row planting. This is the planting of seeds or crops in lines. 	term planting. 2. Mentions examples of planting materials. 3. Identifies the methods of planting.	spells, reads writes and demonstrates meaning of words related to planting 2. Reads, internalize and writes texts and questions related to planting.	Discussion Observation Brain storming	planting Mentioning examples of planting materials. Identifies the methods of planting.	Care Awareness Fluency Concern	dem. Garden .	sci. pbk 4 Int. sci. syllabus bk 4	
5	6	Crop growing	Advantages of row planting	Advantages of row planting method ⇒ It makes weeding easy. ⇒ It controls easy	The learner; 1. States the advantages and disadvantages	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words	Guided discovery	Stating the advantage and disadvantages of row	Appreciation. Care	The school dem. Garden .	Mk. Int. sci. pbk 4	

			method spread of diseases and pests. ⇒ It makes harvesting easy. ⇒ It prevents wastage of seeds and other planting materials. ⇒ It allows proper spacing of crops. Disadvantages of raw planting ⇒ It needs a lot of labour. ⇒ It allows proper spacing of crops. Examples of crops planted by raw planting 1. Maize 2. Pineapples 3. Beans 4. Potatoes 5. Cassava	of raw planting. 2. Mentions examples of crops that can be planted in rows.	related to row planting 2. Reads, internalize and writes texts and questions related to row planting	Discussion Observation Brain storming	planting. Mentioning examples of crops that can be planted in rows.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
5	7	Crop growing	Broadcasting - This is the planting of seeds by throwing them using the hand in a garden.	The learner; 1. The learner describes broadcasting method. 2. States the advantages	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to broadcasting	Guided discovery Discussion	Describing the advantage and disadvantages of broadcasting	Appreciation. Care Awareness	The school dem. Garden .	Mk. Int. sci. pbk 4	

				Illustration  Crops planted by broadcasting method 1. Simsim 2. Millet Advantages of broadcasting method ⇒ It saves time. ⇒ It does not need a lot of labour. Disadvantages of broadcasting method ⇒ It can lead to easy spread of diseases. ⇒ Many times, crops do not yield as expected.	and disadvantages of broadcasting method. 3. Names the examples of crops that can be planted by broadcasting.	method. 2. Reads, internalize and writes texts and questions related to broadcasting method.	Observation Brain storming	ng method. Stating the advantage and disadvantages of broadcasting method. Naming examples crops that can be broadcasted.	Fluency Concern		Int. sci. syllabus bk 4	
6	1	Crop growing	A nursery bed	A nursery bed ⇒ A nursery bed is a place where seedlings are raised.	1. Defines a nursery bed. 2. Mentions examples of crops that can	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words	Guided discovery	Defining a nursery bed.	Appreciation. Care	The school dem. Garden .	Mk. Int. sci. pbk 4	

				Crops that can be raised on a nursery bed Coffee, onions Structure of a nursery bed Importance of a nursery bed ⇒ It protects seedlings from too much direct sunshine. ⇒ It protects seedlings from too much wind and rainfall. Transplanting ⇒ Transplanting is the transfer of seedlings from a nursery bed to the main garden. ⇒ Transplanting is best done in the evening because there is little loss of water through transpiration.	be grown on a nursery bed. 3. States the importance of a nursery bed to the seedlings and the farmer.	related to a nursery bed. 2. Reads, internalize and writes texts and questions related to a nursery bed.	Discussion Observation Brain storming	Mentioning examples of crops that can be raised on a nursery bed. States the importance of a nursery bed.	Awareness Fluency Concern	A chart showing A nursery bed.	Int. sci. syllabus bk 4	
6	2	Crop growing	Ways of caring	Ways of caring for plants Weeding	The learner; 1. Mentions the different ways of	The learner; 1. Pronounces, spells, reads writes	Guided discovery	Mentioning different ways of	Appreciation.	The school dem.	Mk. Int. sci.	

			for plants	Staking Spraying Pruning Thinning Staking ⇒ Staking is the giving of extra support to the plants with stems. ⇒ Staking can be done on tomatoes.  Examples of crops that can be staked 1. Coffee 2. Tomatoes	caring for crops. 2. Describes the meaning of staking and propping, 4. Names examples of crops that can be staked or propped.	and demonstrates meaning of words related caring for crops. 2. Reads, internalize and writes texts and questions related to care for crops.	Discussion Observation Brain storming	caring for crops. Describing the meaning of staking. Naming examples of crops that can be staked.	Care Awareness Fluency Concern	Garden . A chart showing A nursery bed.	pbk 4 Int. sci. syllabus bk 4	
6	3	Crop growing	Weeding	Weeding a) Weeding is the removal of plants from an area where they are not wanted.	The learner; 1. Defines weeding. 2. Mentions examples of weeds.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to	Guided discovery	Mentioning different ways of caring for crops.	Appreciation. Care	The school dem. Garden .	Mk. Int. sci. pbk 4	

				Examples of weeds - nut grass - couch grass Dangers of weeds - They compete with crops for water and mineral salts - Weeds are hiding places for crop pests. Uses of weeds ⇒ For feeding some farm animals ⇒ For thatching houses Ways of controlling weeds. ⇒ By uprooting and burning them ⇒ By mulching Gap filling ⇒ Gap filling is the planting of seeds or seedlings in places where they did not germinate.	3. Identifies garden tools used for weeding. 4. Mentions dangers of weeds.	weeding. 2. Reads, internalize and writes texts and questions related to weeding	Discussion Observation Brain storming	Describing the meaning of staking. Naming examples of crops that can be staked.	Awareness Fluency Concern	A chart showing A weeding tool.	Int. sci. syllabus bk 4	
6	4	Crop growing	Thinning and pruning	Thinning and ⇒ Thinning is the removal of excess seedlings in the garden. Advantages of	The learner; 1. Defines thinning and pruning. 2. States the advantages of thinning and	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to thinning	Guided discovery	Mentioning different ways of caring for crops.	Appreciation. Care	The school dem. Garden .	Mk. Int. sci. pbk 4	

				thinning crops ⇒ It creates space for crops to grow ⇒ It makes weeding easy Commonly thinned crops 1. Cotton 2. Sunflower Pruning ⇒ Pruning is the removal of unwanted parts of a plant. Garden tools used for pruning 1. secateurs 2. pruning saw Reasons why crop farmers prune their crops ⇒ To remove hiding places for crop pest Plants which are pruned include 1. orange plant 2. banana plant	pruning.	and pruning 2. Reads, internalize and writes texts and questions related to thinning and pruning	Discussion Observation Brain storming	Describing the meaning of staking. Naming examples of crops that can be staked.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
6	5	Crop growing	Mulching	Mulching ⇒ Mulching is the covering of top soil with dry plant materials in the	The learner; 1. Defines mulching. 2. States the advantages	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words	Guided discovery	Defining mulching. Stating the advantage	Appreciation. Care	The school dem. Garden .	Mk. Int. sci. pbk 4	

				garden. Examples of mulches Dry banana leaves Advantages of mulching ⇒ Mulching keeps moisture in the soil ⇒ Mulching controls soil erosion Disadvantages of mulching ⇒ Mulches can easily catch fire and burn crops. ⇒ Mulches are hiding places for crop pests e.g. rats. Manuring ⇒ Manuring is the putting of fertilizers in soil to make it more fertile. ⇒ Or manuring is the act of adding fertilizers into the soil. Importance of manure 1. Manure makes the soil more fertile.	and disadvantages of mulches. 3. Describes manuring. 4. States the importance of manuring.	related to mulching and manuring 2. Reads, internalize and writes texts and questions related to mulching and manuring.	Discussion Observation Brain storming	and disadvantages of mulching. Describing manure. Stating the importance of manuring.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
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				2. Manure improves on the expected crop yields.								
6	7	Crop growing	CROP HARVESTING	CROP HARVESTING ⇒ Harvesting is the collection of mature crops from the garden Methods of crop harvesting ⇒ By uprooting using hands e.g. soya beans, beans, ground nuts Storage ⇒ This is the keeping of surplus food safely after harvesting ⇒ Seeds and cereals after sun drying them, should be stored properly A storage facility (granary) rat guards prevent rats from entering	The learner; 1. Defines harvesting. 2. Mentions garden tools for harvesting different crops. 3. Identifies ways /methods of harvesting	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to harvesting. 2. Reads, internalize and writes texts and questions related to care for crops.	Guided discovery Discussion Observation Brain storming	Defining the term harvesting. Mentioning garden tools for harvesting. Identifying ways/ methods of harvesting	Appreciation. Care Awareness Fluency Concern	The school dem. Garden .	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				into the granary Reasons for storing food ⇒ To prevent wastage of food ⇒ To sell it when there is good market Conditions for proper storage of food ⇒ The seeds or grains should be stored when they are dry									
7	1	Crop growing		Record keeping ⇒ This is a practice where a farmer writes down all the activities done on the farm. ⇒ Farm records are written information about various activities carried out on a farm. Types of farm records ⇒ production records ⇒ inventory records Reasons why crop	The learner; 1. Defines record keeping. 2. States the common records kept on a crop farmer. 3. Gives the importance of keeping records.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to record keeping. 2. Reads, internalize and writes texts and questions related to record keeping.	Guided discovery Discussion Observation Brain	Defining the term harvesting. Mentioning garden tools for harvesting. Identifying ways/ methods of harvesting	Appreciation. Care Awareness Fluency Concern	The school dem. Garden .	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4		

				farmers keep records ⇒ They enable a crop farmer to calculate profits and losses ⇒ They enable a crop farmer to plan for his farm			storming					
7	2	Crop growing	Crop pests	Crop pests ⇒ A crop pest is a living organism which destroys crops. ⇒ They include: animals, birds, and insects. Dangers of crop pests ⇒ They reduce crop yields ⇒ They lower the quality of crop yields Signs of pest damage on crops ⇒ Some leaves are partly eaten up or have holes ⇒ Fruits develop dark spots ⇒ Cut off buds ⇒ Seeds with holes	The learner; 1. Defines crop pests. 2. States the Dangers of crop pests. 3. Mentions signs of crop pests in the garden	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to crop pests. 2. Reads, internalize and writes texts and questions related to crop pests.	Guided discovery Discussion Observation Brain storming	Defining crop pests. Stating the dangers of crop pests. Mentioning signs of crop pests in the garden.	Appreciation. Care Awareness Fluency Concern	The school dem. Garden .	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				⇒Rotten tubers ⇒Stunted growth.								
7	3	Crop Growi ng	Crop pests for differe d crops.	Pests for different crop pests Legumes - bean weevils - Cut worm - Bean fly - thrips - termites tubers - rats - mice - Caterpillars Cereals - locusts - monkeys - maize weevils How to control crop pests ⇒By spraying crops with pesticides ⇒By putting scare crows in the garden. Crop diseases - Rosette - Tomato blight - Maize streak	The learner; 1. Identifies crop pests for different crops. 2. States the ways of controls crop pests. 3. Mentions the different crop diseases.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to crop pests. 2. Reads, internalize and writes texts and questions related to crop pests.	Guided discovery Discussion Observati on Brain storming	Identifying crop pests for different crops. Stating the ways of controlling crop pests. Mentions the different crop diseases.	Appreciati- on. Care Awareness Fluency Concern	The school dem. Garden .	Mk. Int. sci. pbk 4 Int. sci. syllab us bk 4	
7	4	Weath er chang	Types of weath	Weather, What is weather? Weather is the state	The learner; 1. Give the meaning of	The learner; 1. Pronounces, spells, reads	Guided discovery	Giving the meaning of weather	Appreciati- on.	The school dem.	Mk. Int. sci.	

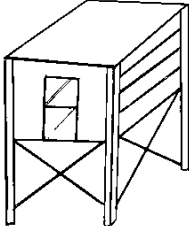
		es around us	er	of the atmosphere at a given time and place. Types of weather. Elements / factors of weather. - Sunshine - Rainfall - Cloud Cover - Wind - Humidity - Temperature Rainfall ⇒ Rainfall is water falling in separate drops from clouds. ⇒ Rainfall is the amount of rain water that falls in a certain area at a certain time.	weather. 2. Identifies the types of weather. 3. states the weather elements	writes and demonstrates meaning of words related to weather 2. Reads, internalize and writes texts and questions related to weather	Discussion Observation Brain storming	Identifying the types of weather. Stating the elements of weather.	Care Awareness Fluency Concern	Garden .	pbk 4 Int. sci. syllabus bk 4	
7	5	Weath er chang es around us	The water cycle	WATER CYCLE (rain cycle) How rain is formed. - The sun heats water in water bodies and plants to produce water vapour. - Water vapour rises into the atmosphere.	The learner; 1. Describes the water cycle. 2. Differentiates between rain and rainfall. 3. Draws the structure of the water cycle.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to weather 2. Reads, internalize and	Guided discovery Discussion	Describing the water cycle. Differentiating between a rain and rain fall.	Appreciation. Care Awareness	The school dem. Garden .	Mk. Int. sci. pbk 4 Int. sci. syllabus	

				- Water vapour condenses to form clouds. - When the clouds become heavy, they form rain. An illustration of a water cycle. Process A – Transpiration B – Evaporation C – Condensation		writes texts and questions related to weather	Observation Brain storming	Drawing the structure of the water cycle.	Fluency Concern		us bk 4	
7	6	Weather changes around us	The water cycle	An experiment to show water cycle. Things needed - A kettle of water - Burning charcoal - Iced bottle	The learner; 1. Describes an experiment to show the water cycle. 2. Identifies the processes involved in the experiment.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to water cycle 2. Reads, internalize and writes texts and questions related to water cycle.	Guided discovery Discussion Observation Brain storming	Describing an experiment on the water cycle. Identifying the process involved in the water cycle.	Appreciation. Care Awareness Fluency Concern	A chart showing an illustration of the water cycle. 	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
7	7	Weather	How to measure	How to measure rainfall.	The learners; 1. Describes the	The learner; 1. Pronounces,	Guided discovery	Describing the way	Appreciation.	A chart showing	Mk. Int.	

		changes around us.	re rainfall	- Rainfall is measured by an instrument called a rain gauge. - Rainfall is measured in millimetres. A DIAGRAM OF A RAIN GAUGE Importance of rainfall. - Rainfall cools the temperature. - It provides water for seeds. Dangers of too much rainfall. - Heavy rainfall causes floods that can destroy the environment. - Floods carry rubbish and soil into lakes and rivers.	way in which rainfall is measured. 2. Draws the structure of the rain gauge. 3. Gives the importance of rain to people. 4. States the danger of rainfall.	spells, reads writes and demonstrates meaning of words related to water cycle 2. Reads, internalize and writes texts and questions related to water cycle.	Discussion Observation Brain storming	rainfall is formed. Drawing the diagram of the rain gauge. Giving the importance of rain. Stating the dangers of rainfall.	Care Awareness Fluency Concern	g an illustration of the water cycle. Int. sci. syllabus bk 4	
8	1	Weather changes around us	Clouds	Clouds - Clouds are a mass of condensed water vapour. Types of clouds Cirrus - furthest in the sky. Cumulus	The learners; 1. Gives the meaning of clouds. 2. Identifies the types of clouds. 3. Gives the	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to weather chart	Guided discovery Discussion	Giving the meaning of clouds. Identifying the types of clouds.	Appreciation. Care Awareness	A chart showing an illustration of the water cycle. Mk. Int. sci. pbk 4 Int.	

				Stratus Nimbus - lowest clouds Importance of clouds. - They give us rain. - They modify weather (regulate temperature) Temperature: -Temperature is the degree of hotness or coldness of a body or a place. - The instrument used to measure temperature is called a thermometer. - Types of thermometers. - Clinical thermometer - six's thermometer - wall thermometer	importance of clouds. 4. Defines temperature. 5. Describes the types of thermometers.	2. Reads, internalize and writes texts and questions related to weather chart	Observation Brain storming	Giving the importance of clouds. Defining temperature. Describing the types of thermometers.	Fluency Concern		sci. syllabus bk 4	
8	2	Weath er changes around us	Wind	Wind ⇒ Wind is moving air Wind instruments: a). Wind vane ⇒ It shows the direction where wind is blowing from. b). Wind sock	The learner. 1. Describes wind. 2. State the role of a wind vane. 3. Draws the structure of a wind vane, wind sock, and anemometer.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to wind 2. Reads,	Guided discovery Discussion	Giving the meaning of wind Stating the instrument use in windy weather.	Appreciation. Care Awareness	An illustration of a wind vane, wind sock and anemo	Mk. Int. sci. pbk 4 Int. sci.	

				⇒ A wind sock and weather cock can show the direction of wind. c). Anemometer ⇒ It is used to measure the speed of wind. ⇒ it has cups which trap wind and rotate as wind blows. Uses of wind: ⇒ Wind is used for winnowing: ⇒ Wind brings cold air in warm places Disadvantages of wind: ⇒ Strong wind leads to soil erosion. ⇒ Strong wind destroys property.	4. States the advantages and disadvantages of wind of wind.	internalize and writes texts and questions related to wind	Observation Brain storming	Drawing the structure of a wind vane, wind sock and anemometer Stating the advantage and disadvantages of wind.	Fluency Concern	meter.	syllabus bk 4	
8	3			<u>WEATHER STATION</u> ⇒ What is a weather station? ⇒ A weather station is a place where weather conditions are studied.	The learner; 1. Defines a weather station and weather forecast. 2. Draws a Stevenson screen	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to a weather	Guided discovery Discussion	Giving the meaning of a weather station and weather forecast.	Appreciation. Care Awareness	An illustration of a Stevenson screen	Mk. Int. sci. pbk 4 Int.	

				Weather forecast: ⇒ This is the prediction of future weather changes. <u>The structure of a Stevenson screen</u>  ❖ NB: The Stevenson screen is painted white so as to reflect heat. ❖ It is used to keep delicate weather instruments like: 1. Thermometers 2. Barometers 3. Hygrometers	3. Identifies the weather instrument kept in a Stevenson screen.	station 2. Reads, internalize and writes texts and questions related to weather forecast.	Observation Brain storming	Drawing the structure of a Stevenson screen Identifying the instruments kept in a Stevenson screen.	Fluency Concern		sci. syllabus bk 4	
8	4	PERSONAL HYGIENE	PERSONAL HYGIENE	PERSONAL HYGIENE ⇒ Personal hygiene is the general cleanliness of the body. ⇒ Or personal hygiene is the keeping of the	The learner; 1. Defines personal hygiene. 2. States ways of keeping the body clean. 3. Identifies the	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to Personal	Guided discovery Discussion	Defining personal hygiene. Stating ways of keeping the body	Appreciation. Care Awareness	Clean water A tooth brush	Mk. Int. sci. pbk 4	Int.

				body clean. Ways of keeping the body clean - Bathing daily. - Cutting finger nails short. - Washing hands after using a latrine/ toilet. - Washing clothes. - Combing hair. - Ironing clothes. Items used in keeping our bodies clean - Clean water - Tooth paste tooth brush - Bathing sponge - Towel - dental floss	items used to keep the body clean.	hygiene Reads, internalize and writes texts and questions related to Personal hygiene	Observation Brain storming	clean. Identifying the items used to keep the body clean.	Fluency Concern	Tooth paste	sci. syllabus bk 4	
8	5	Personal hygiene	Importance of keeping the body clean	Importance of keeping the body clean ⇒ It helps to prevent skin diseases. ⇒ It helps to prevent bad smell caused by sweating. Effects of poor personal hygiene ⇒ It leads to bad body smell.	The learner; - States the importance of keeping our bodies clean. - States the dangers of poor personal hygiene.	The learner; - Pronounces, spells, reads writes and demonstrates meaning of words related to Personal hygiene - Reads, internalize and writes texts and	Guided discovery Discussion Observation	Stating the importance of keeping our bodies clean. Stating the dangers of poor personal	Appreciation. Care Awareness Fluency	Clean water A tooth brush Tooth paste	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				⇒ It leads to skin diseases. Keeping clothing and beddings clean ⇒ Beddings should be washed regularly with clean water and soap. ⇒ Clothes need to be ironed after washing in order to kill parasites and germs. Diagram of a child ironing • Importance of keeping beddings clean. ⇒ It prevents bad smell. ⇒ It prevents parasites like lice, ticks and fleas.		questions related to Personal hygiene	on	hygiene	Concern			
							Brain storming					

PRIMARY FOUR SCIENCE SCHEME OF WORK FOR TERM III, 2025

W K	P D	TOPIC	TOPIC	CONTENT	SUBJ. COMP'NCES	LANG. COMP'NCES	METHODS	ACTIVITIES	IND. OF L.S.V	INST. MAT.	REF.	R E M
1	1	comm unicab le intestin al diseas es and worm infesta tions	Comm unicab le Diseas es	<u>COMMUNICABLE DISEASES;</u> These are diseases which can be spread from one person to another. <u>Diarrheal diseases</u> These are diseases that affect the gut, and they are characterized by diarrhea Examples include; 1. Typhoid 2. Dysentery 3. Peptic ulcers	The learner; -Defines the term communicable diseases -Names different communicable diseases -States different causes of intestinal communicable diseases	The learner; -Pronounces, spells, reads, writes and demonstrates, words related to intestinal communicable diseases	-Class discussion -Guided discovery -Brain storming	Naming examples of diarrhea diseases	Awareness, Care, appreciatio n	work cards showin g exampl es of diarrhe a disease s	Mk inte grat ed scie nce	1
2	2	comm unicab le intestin al diseas es and worm infesta tions	How intestin al diseas es are spread	<u>How intestinal diseases are spread.</u> 1. Through eating contaminated food. 2. Through drinking contaminated water. 3. Through a 4 F path germ	The learner; describes the 4Fs path germ	The learner; Pronounces, spells, reads, and writes and demonstrated the meaning of words related to 4Fs path germ	-Brain storming, -Guided discovery	Describing the 4Fs path germ.	Fluency Observatio n care awareness	A chart showin g 4Fs path germ.	Co mpr ehe nsiv e scie nce boo k 4.	

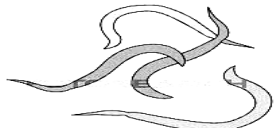
		Comm unicab le intestin al diseas es and worm infesta tions	DIARR HOEA	<u>DIARRHOEA</u> - This is a disease caused by either a virus or bacteria or amoeba. - Diarrhea involves passing out of watery stools frequently. <u>Effects of diarrhea</u> -Diarrhea causes dehydration	The learner; - defines diarrhea. -states the ways through which diarrhea is spread.	The learner; Pronounces, spells, reads, and writes and demonstrated the meaning of words related to diarrhea	Class discussion -Guided discovery -Brain Storming.	Identifying the ways through which diarrhea is spread.	Fluency, appreciation, care.	Work cards showing the spread of diarrhea	Mk integrated science pupils book 4	2
2	4	Comm unicab le intestin al diseas es and worm infesta tions	CHOLE RA AND TYPHOID	<u>DYSENTERY</u> -This is diarrhea with blood or mucus. -It is caused by either bacteria or amoeba. signs and symptoms of dysentery -pain in the abdomen -bloody diarrhea -vomiting CHOLERA It is caused by bacteria Signs and symptoms of Cholera -watery stool -frequent vomiting -dehydration	The learner; -States the cause of cholera and typhoid -Gives signs and symptoms of cholera and typhoid.	The learner; Pronounces, spells, reads, and demonstrated the meaning of the words related to cholera and typhoid.	Class discussion Guided discovery -Brain Storming	Identifying the ways through which typhoid and cholera are spread	Fluency, appreciation, care.	Work cards showing the spread of cholera and typhoid	Mk integrated science pupils book 4	2
1	5	Comm unicab le	Preven tion, control	Prevention, control and treatment of diarrhoeal diseases.	The learner; -States the cause of typhoid	The learner; Pronounces, spells, reads, and	Class discussion Guided	Stating	Fluency, appreciation, care.	Work cards showing	Mk integrated	

		intestinal diseases and worm infestations	and treatment of diarrhoeal diseases.	It is caused by bacteria Signs and symptoms of Typhoid ⇒ Abdominal pain ⇒ Fever ⇒ headaches Effects of typhoid ⇒ diarrhea ⇒ dehydration ⇒ death The 3 Ds ⇒ Diarrhea ⇒ Dehydration ⇒ Death	-Gives signs and symptoms of typhoid.	demonstrated the meaning of the words related to typhoid.	discovery -Brain Storming			g the spread of cholera and typhoid	ed science pupils book 4	
1	6	Communicable intestinal diseases and worm infestations	Prevention, control and treatment of diarrhoeal diseases.	Prevention, control and treatment of diarrhoeal diseases. ⇒ Washing hands with soap and clean water. ⇒ Cleaning of toilets Effect of diarrhoeal diseases: They can lead to; Dehydration ⇒ Dehydration is the condition when the body does not have enough water. Causes of dehydration	The learner, 1. States the ways of preventing the spread of diarrhoeal diseases. 2. States the effect of diarrhoeal 3. States the causes of dehydration. 4. Mentions signs and symptoms of dehydration.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to prevention and control of Diarrhoeal diseases 2. Reads, internalize and writes texts and questions related to prevention and control of	Guided discovery Discussion Observation illustration	Stating the ways of preventing the spread of diarrhoeal diseases. Stating the effect of diarrhoeal Stating the causes of dehydration · Mentioning	Appreciation. Care Awareness Fluency Concern	A picture of a dehydrated child.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				⇒ Too much diarrhoea ⇒ Too much vomiting Signs and symptoms of dehydration. ⇒ Sunken eyes. ⇒ Little or no urine or tears.		Diarrhoeal diseases		signs and symptoms of dehydration . plant.				
2	1	Communicable intestinal diseases and worm infestations	Treatment of dehydration.	Treatment of dehydration. ⇒ The patient should be given Oral Rehydration solution (ORS) ⇒ ORS is given in order to replace the lost body fluids. Preparation of ORS / SSS – Salt sugar solution ORS – from a sachet. What is needed? 1. One sachet of ORS (oral rehydration salts) 2. 1 litre (2 tumpecos) of drinking water. 3. Spoon 4. Water 5. Clean container	The learner; 1. Mentions the treatment given to a dehydrated person. 2. Illustrate how to prepare ORS/ SSS	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to treatment of dehydration 2. Reads, internalize and writes texts and questions related to treatment of dehydration.	Guided discovery Discussion Observation illustration	Mentioning the treatment given to a dehydrated person. Illustrating how to prepare ORS/ SSS	Appreciation. Care Awareness Fluency Concern	Salt Sugar Water water jar	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

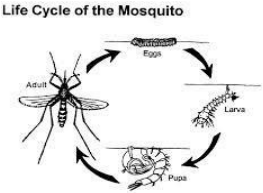
				Steps taken to prepare ⇒ Wash your hands with water and soap.								
2	2	Communicable intestinal diseases and worm infestations	Preparation of S.S.S – Salt Sugar solution Requirements	Preparation of S.S.S – Salt Sugar solution Requirements ⇒ water and soap ⇒ clean boiled cool water Steps taken ⇒ Wash your hands with water and soap. ⇒ Measure one litre of clean cool boiled water and pour it in a clean container. ⇒ Measure 8 levelled tea spoons of sugar and one leveled tea spoon of salt into the container	The learner; 1. Identifies the steps involved in preparing ORS/SSS. 2. Practically prepares ORS/ s.s.s	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to preparation of ORS/SSS 2. Reads, internalize and writes texts and questions related to preparation of ORS/ SSS	Guided discovery Discussion Observation Illustration Demonstration.	Identifying the steps involved in preparing ORS/SSS. Practically preparing ORS/ s.s.s	Appreciation. Care Awareness Fluency Concern	Salt Sugar Water water jar	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
2	3	Communicable intestinal diseases and	WORM INFESTATION	WORM INFESTATION ⇒ Worms are long, slender soft bodied animals with no limbs. ⇒ Worms can live in soil, bodies of	The learner; 1. Describes worm infestations. 2. Mentions the examples of worms.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to	Guided discovery Discussion Observation	describing worm infestations Mentioning the examples of	Appreciation. Care	A chart showing different types of worms.	Mk. Int. sci. pbk 4	

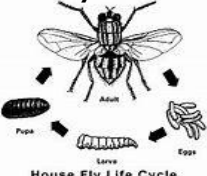
		worm infestations		animals, in fresh water, in plants, fruits or vegetables. Examples of intestinal worms ⇒ tape worms ⇒ hook worms How we can get worms. ⇒ Eating contaminated food with worms or eggs ⇒ Drinking contaminated water-Eating under cooked meat (pork/ beef)	3. States how worms get into our bodies.	worm infestations. 2. Reads, internalize and writes texts and questions related to worm infestations.	on Illustration Demonstration.	worms. Stating how worms gets into our bodies.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
2	4	Communicable intestinal diseases and worm infestations	Signs and symptoms of worm infestation	Signs and symptoms of worm infestation ⇒ diarrhoea ⇒ loss of weight TAPE WORMS ⇒ They are flat ⇒ They have segments (segmented) Structure of a tape worm. Note:	The learner; 1. Describes signs and symptoms of tape worm infestation 2. Describes the structure of a tape worms 3. States how tape worms get into our bodies.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to tape worm infestations. 2. Reads, internalize and writes texts and questions	Guided discovery Discussion Observation Illustration	Describing the signs and symptoms of tape worm infestations Stating how tapes worms get into our bodies. Stating how	Appreciation. Care Awareness	A chart showing a structure of a tape worm.	Mk. Int. sci. pbk 4 Int. sci. syllabus	

				⇒The head of a tape worm is called a Scolex ⇒The hooks and suckers help tapeworms to hold on to the intestines Prevention of tape worms. ⇒Keeping toilets and latrines clean ⇒Proper disposal of faeces in latrines Treatment of tape worms. ⇒ Deworming all family members and animals at home regularly.	4. States the ways of preventing and treating tape worm infestations.	related to tape worm infestations.	Demonstration.	ways of controlling the spread of tape worm infestations	Fluency Concern		bus bk 4	
2	5	Communicable intestinal diseases and worm infestations	HOOK WORMS	HOOK WORMS ⇒ They feed on blood ⇒ They live in small intestines Structure of a hook worm.  Signs and symptoms of hookworms	The learner; 1. Describes the structure of a hook worm. 2. Describes signs and symptoms of a hook worm infestation. 3. States how tapeworms get into our bodies.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to hook worm infestations. 2. Reads, internalize and writes texts and questions	Guided discovery Discussion Observation Illustration	Describing the signs and symptoms of hook worm infestations Stating how hook worms get into our bodies. Stating how	Appreciation. Care Awareness	A chart showing a structure of a hook worm.	Mk. Int. sci. pbk 4 Int. sci. sylla	

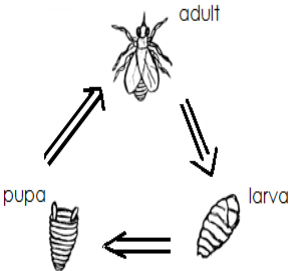
				⇒ A person becomes anaemic ⇒ pale skin, pale palm, pale eyelids Prevention and treatment of hookworm infestation. ⇒ wear shoes or sandals especially in dirty places ⇒ Always defecate in the latrine or toilet but not in bushes	4. States the ways of preventing and treating tapeworm infestations.	related to hook worm infestations.	Demonstration.	ways of controlling the spread of hook worm infestations	Fluency Concern		bus bk 4	
2	6	Communicable intestinal diseases and worm infestations	PIN WORMS (Thread worms)	⇒ They live in the colon and rectum. ⇒ They lay their eggs around the anus. ⇒ They cause itching around the anus at night. Structure of pin worms. Signs and symptoms ⇒ The person passes out the worms in the faeces.	The learner; 1. Describes signs and symptoms of pin infestation 2. Describes the structure of a tape 3. States how pin worms get into our bodies. 4. States the ways of preventing and treating pin	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to pin worm infestations. 2. Reads, internalize and writes texts and questions related to pin worm infestations.	Guided discovery Discussion Observation Illustration Demonstration	Describing the signs and symptoms of pinworm infestations Stating how pin worms get into our bodies. Stating how ways of controlling the spread	Appreciation. Care Awareness Fluency	A chart showing a structure of pin worms worm.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				⇒ Swollen belly. ⇒ Itching anus Prevention and treatment of thread worm infestation ⇒ Wash your hands before handling any food and after using the toilet. ⇒ Practice proper personal and food hygiene	worm infestations.		ation.	of pin worm infestations.	Concern			
3	1	Vectors and diseases	Common vectors.	Common vectors. A vector is an organism that spreads germs. Examples of common vectors  Characteristics of vectors ⇒ Some vectors have a proboscis for sucking blood e.g. mosquitoes, ticks, fleas, lice and tsetse flies. ⇒ Some have a hairy body for carrying germs	The learner; 1. Describes a vector. 2. States examples of vectors.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to vectors. 2. Reads, internalize and writes texts and questions related to vectors	Guided discovery Discussion Observation Illustration Demonstration.	Describing vector. Stating examples of vectors.	Appreciation. Care Awareness Fluency Concern	A chart showing common vectors in the environment	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				e.g. a house fly								
3	2	Vector s and diseas es	How vector s spread diseas es. Mosqui toes	How vectors spread diseases. Mosquitoes Mosquitoes lay their eggs in stagnant water (still water) Life cycle of a mosquito  How mosquitoes spread germs; ⇒ Germs are spread by an adult mosquito through bites. Diseases spread by different mosquitoes. ⇒ Female anopheles mosquito – malaria ⇒ Tiger / Aedes mosquito – yellow fever	The learner; 1. States how mosquitoes spread diseases. 2. Describes the life cycle of a mosquito 3. States how mosquitoes spread diseases. 4. Mentions the diseases spread by mosquitoes.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to how mosquitoes spread diseases 2. Reads, internalize and writes texts and questions related to how mosquitoes spread diseases.	Guided discovery Discussion Observation Illustration Demonstration.	Stating how mosquitoes spread diseases. Describing the life cycle of a mosquito Stating how mosquitoes spreads diseases Mentioning the diseases spread by mosquitoes	Appreciation. Care Awareness Fluency Concern	A chart showing a life cycle of a mosquito. Int. sci. syllabus bk 4	Mk. Int. sci. pbk 4	
3	3	Vector	House	House flies	The learner;	The learner;	Guided	Stating how	Appreciation-	A chart	Mk.	

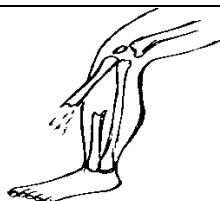
			s and diseases	flies	⇒ Houseflies live in dirty places e.g. latrines and rubbish pits. Lifecycle of a housefly  House Fly Life Cycle ⇒ Houseflies lay eggs in decaying matter for their larvae to feed on decayed matter. How houseflies spread germs. ⇒ Germs are spread by an adult housefly using its hairy body when it visits dirty places. ⇒ The germs are carried onto our uncovered food. Diseases spread by houseflies diarrhoea typhoid dysentery cholera	1. States how house flies spread diseases. 2. describes the life cycle of a house fly 3. States how house flies. Spread diseases. 3. Mentions the diseases spread by a housefly	1. Pronounces, spells, reads writes and demonstrates meaning of words related to how houseflies spread diseases 2. Reads, internalize and writes texts and questions related to how houseflies spread diseases.	discovery Discussion Observation Illustration Demonstration.	houseflies spread diseases. Describing the life cycle of a housefly Stating how a house fly Spreads diseases. Mentioning the diseases spread by houseflies.	on. Care Awareness Fluency Concern	showing a life cycle of a housefly	Int. sci. pbk 4 Int. sci. syllabus bk 4	
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3	4	Vector s and diseas es	Cockr oache s	Cockroaches - Cockroaches are commonly found in dark corners and dark places e.g. cupboards, wall cracks and latrines. Lifecycle of a cockroach. How cockroaches spread germs. ⇒ Cockroaches pick germs from dirty places which they carry on their hairy body. ⇒ When they land on uncovered food, they leave germs on it and we get them when we eat that food. Diseases spread by cockroaches. ⇒ Diarrhoea ⇒ Cholera ⇒ Dysentery ⇒ Typhoid	The learner; 1. States how cockroaches spread diseases. 2. describes the life cycle of a cockroach 3. States how cockroaches spread diseases. 4. Mentions the diseases spread by a cockroach	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to how cockroaches spread diseases 2. Reads, internalize and writes texts and questions related to how cockroach spread diseases.	Guided discovery Discussion Observation Illustration Demonstration.	Stating how cockroaches spread diseases. Describing the life cycle of a cockroach Stating how house fly spread diseases. Mentioning the diseases spread by cockroaches	Appreciation. Care Awareness Fluency Concern	A chart showing a life cycle of a cockroach	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
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3	5	Vector s and diseas es	Lifecyc le of a tsetse fly	Lifecycle of a tsetse fly  An adult housefly doesn't lay eggs, the eggs hatch inside the body and passes out the larvae. How a tsetse fly spreads diseases ⇒ By biting Disease spread by tsetse flies ⇒ Sleeping sickness (to people) ⇒ Nagana (to animals) Other diseases vectors rat fleas lice	The learner; 1. States how tsetse flies spread diseases. 2. describes the life cycle of a tsetse flies 3. States how a tsetse fly spreads diseases. 4. Mentions the diseases spread by tsetse flies	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to how tsetse flies 2. Reads, internalize and writes texts and questions related to tsetse flies	Guided discovery Discussion Observation Illustration Demonstration.	Stating how cockroaches spread diseases. Describing the life cycle of a cockroach Stating how tsetse fly spreads diseases. Mentioning the diseases spread by tsetse flies	Appreciation. Care Awareness Fluency Concern	A chart showing a life cycle of a tsetse flies.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
3	6	Vector s and diseas es	Preven tion and control	Prevention and control of disease vectors and diseases. Ways of controlling	The learner; 1. States ways of controlling the spread of	The learner; 1. Pronounces, spells, reads writes and	Guided discovery Discussion	Stating ways of controlling the spread	Appreciation. Care	The environment	Mk. Int. sci. pbk	

		and first aid		sudden happening that may cause harm to the body. ⇒ A person involved in an accident is called a casualty. Examples of accidents at home ⇒ Cuts and scalds ⇒ Poisoning ⇒ Drowning ⇒ Electric shocks Accidents on the way to school ⇒ motor accidents ⇒ stings ⇒ cuts ⇒ bites	accident. 2. Mentions examples of accidents at home and at school.	spells, reads writes and demonstrates meaning of words related to accidents. 2. Reads, internalize and writes texts and questions related to accidents.	Discussion Observation Illustration Demonstration.	Mentioning examples of accidents at home and at school.	Care Awareness Fluency Concern	ment	sci. pbk 4 Int. sci. syllabus bk 4	
4	2			Types of injuries and their causes. Burns ⇒ A burn is an injury caused by dry heat. Source causes of burns. ⇒ hot charcoal ⇒ hot metals ⇒ fire	The learner; 1. Defines a burn and a scald. 2. Mentions the sources of burns and scalds. 3. Mentions ways of preventing burns and scalds.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to burns and scalds. 2. Reads, internalize and	Guided discovery Discussion Observation	Defining a burn and a scald. Mentioning the sources of burns and scalds. Mentioning ways of	Appreciation. Care Awareness	The environment Chalk board illustration.	Mk. Int. sci. pbk 4 Int.	

				⇒ electricity Scalds A scald is an injury on the body caused by wet heat. Sources / causes of scalds 1. hot water 2. steam 3. hot porridge 4. hot soup		writes texts and questions related to burns and scalds.	Illustration Demonstration.	preventing burns and scalds.	Fluency Concern		sci. syllabus bk 4	
4	3			Fracture A fracture is a broken or cracked bone in the body. Causes of fractures. ⇒ Falling from a tree, bed, building etc. ⇒ Getting involved in motor accidents ⇒ Playing rough games. Types of fractures. Simple fracture Compound fracture	The learner; 1. Defines a fracture. 2. Mentions the causes of fractures. 3. Describes the types of fractures.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to fractures. 2. Reads, internalize and writes texts and questions related to fractures.	Guided discovery Discussion Observation Illustration Demonstration.	Defining a fracture. Mentioning the causes of fractures. Describing the types of fractures.	Appreciation. Care Awareness Fluency Concern	The environment A chart showing a simple and compound fracture.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	



Simple fracture

⇒ This is when the bone breaks or cracks and it remains inside the skin.

Structure of a simple fracture.

Compound fracture.

⇒ This is where the bone breaks and comes out of the skin.

Structure of a compound fracture.

Sprains, strains and dislocations

⇒ A sprain is a twisted or torn ligament.

⇒ A ligament is a structure that joins a bone to bone.

Causes of sprains.

Twisting or tearing of

The learner;

1. Defines a strain a sprain and a dislocation.

2. Mentions the causes strains, sprain and dislocations.

States the first aid

The learner;

1. Pronounces, spells, reads writes and demonstrates meaning of words related to sprains, strains and dislocations.

Guided discovery

Discussion

Observation

Defining a strain a sprain and a dislocation.

Mentioning the causes strains, sprain and dislocations.

Appreciation.

Care

Awareness

Chalkboard illustration.

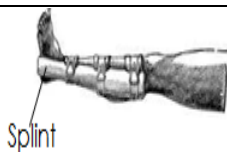
Mk. Int. sci. pbk 4

				the ligament Signs and sprains ⇒ Swelling around the joint. ⇒ Pain in the joint. Stains ⇒ A strain is an injury on the tendon or muscle. ⇒ A tendon is a tissue that joins a muscle to a bone Causes of strains. ⇒ Over stretching of a muscle. Signs of a strain ⇒ Sudden sharp pain in a muscle. ⇒ The muscle may appear swollen. Dislocation ⇒ This is where a bone gets out of its usual position.	for dislocations.	2. Reads, internalize and writes texts and questions related to sprains, strains and dislocations.	Illustration Demonstration.	Stating the first aid for dislocations.	Fluency Concern		Int. sci. syllabus bk 4	
4	5			Ways of preventing accidents How to prevent accidents at home. ⇒ Keep medicine, drugs and other chemicals out of children's reach. ⇒ Cooking should	The learner; 1. States ways of controlling and preventing accidents at and at school.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to prevention of accidents.	Guided discovery Discussion Observation	Sating ways of controlling and preventing accidents at and at school..	Appreciation. Care Awareness	Road traffic signs	Mk. Int. sci. pbk 4	

				be done in a raised place where children cannot reach. ⇒ Avoid playing with electric equipment Preventing accidents on the way to school. ⇒ Avoid on the road. ⇒ Use a zebra crossing when crossing the road.		2. Reads, internalize and writes texts and questions related to prevention of accidents/	Illustration Demonstration.		Fluency Concern		Int. sci. syllabus bk 4	
4	6			Safety rules on the road. ⇒ Look right and then left, look right again before you cross the road. ⇒ Use a zebra crossing to cross the road. ⇒ Avoid playing on the road. ⇒ Always walk on the pedestrian side of the road. Examples of road signs	The learner; 1. States the safety rules on the road. 2. Identifies examples of road signs.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to prevention of accidents. 2. Reads, internalize and writes texts and questions related to prevention of accidents/	Guided discovery Discussion Observation Illustration Demonstration.	Sating the safety rules on the road. Identifying examples of road signs.	Appreciation. Care Awareness Fluency Concern	Road traffic signs	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
5	1			Poisoning	The learner;	The learner;	Guided	Defining	Appreciation-	Jik	Mk.	

				⇒ Poison is anything that can cause harm or death when it gets in the body. ⇒ Poisoning happens when one eats or drinks something that is harmful to the body. Causes of poisoning ⇒ Taking expired drugs. ⇒ Taking over dose of medicine. Signs and symptoms of poisoning. 1. Vomiting 2. Fainting 3. Difficulty or rapid breathing Prevention of poisoning. 1. Keep drugs away from children's reach. 2. Take drugs prescribed by health workers.	1. Defines poison and poisoning. 2. Mentions the causes of poisoning 3. Identifies the signs and symptoms of poisoning. 4. States ways of preventing poisoning.	1. Pronounces, spells, reads writes and demonstrates meaning of words related to poisoning. 2. Reads, internalize and writes texts and questions related to poisoning.	discovery Discussion Observation Illustration Demonstration.	poison and poisoning. Mentioning the causes of poisoning Identifying the signs and symptoms of poisoning. Stating ways of preventing poisoning.	on. Care Awareness Fluency Concern	Vim rat poison Insecticides.	Int. sci. pbk 4 Int. sci. syllabus bk 4	
5	2			First aid Kit A first aid kit is a collection of first aid	The learner; 1. Defines a first aid kit.	The learner; 1. Pronounces, spells, reads	Guided discovery	Defining a first aid kit.	Appreciation.	Jik	Mk. Int. sci.	

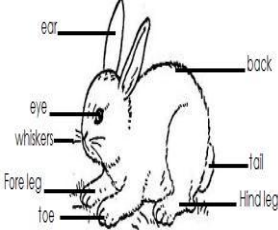
				materials Components of a first aid kit and their uses. A razor blade. A safety pin. A pair of scissors A gauze plaster	2. Identifies the components of a first aid kit with their uses in first Aid administration.	writes and demonstrates meaning of words related to first aid kit 2. Reads, internalize and writes texts and questions related to first aid kit	Discussion Observation Illustration	Identifying the components of a first aid kit with their uses in first aid administration.	Care Awareness Fluency Concern	Vim rat poison Insecticides.	pbk 4 Int. sci. syllabus bk 4	
5	3			Giving first aid ⇒ First aid is the immediate help given to a casualty before being taken to the health centre. Reasons for giving first aid ⇒ -save life. ⇒ To reduce pain. First aid for burns and scalds ⇒ Cool the injured part with clean cold water. First aid for fractures. ⇒ Tie splints on the fractured area. A splint on the leg	The learner; 1. Defines a first aid 2. Gives reasons for giving first aid.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to first aid 2. Reads, internalize and writes texts and questions related to first aid.	Guided discovery Discussion Observation Illustration	Defining a first aid. Giving reasons for giving first aid.	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	



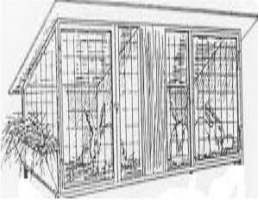
- ⇒ Splints hold the broken bone in one position.
- ⇒ Support the injured part with an **arm sling** in case it is the arm that is fractured.

5	4			First aid for sprains, strains and dislocations. ⇒ Apply cold water or ice at the injured part. Cold water reduces pain and swelling in the tissue. ⇒ Fix the injured part with a bandage. First aid for poisoning. ⇒ Give the person milk, juice or water to take in order to dilute the poison. ⇒ Make the person vomit in case of	The learner; 1. Describe and illustrates the first aid for strains, sprains and poisoning.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to first aid 2. Reads, internalize and writes texts and questions related to first aid.	Guided discovery Discussion Observation Illustration	Describing the first aid for strains, sprains and poisoning.	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				solid poison.								
5	5			First aid for snake bites and nose bleeding. First aid for snake bites. ⇒ Tie a tourniquet between the bitten part and the heart to prevent venom (poison from reaching the heart.) ⇒ A tourniquet is a piece of cloth used to tie near the bitten part. First aid for nose bleeding ⇒ Bend the patients head forward. ⇒ Pinch the nose of the patient and advise the patient to breathe through the mouth.	The learner; 1. Describe and illustrates the first aid for snake bites and nose bleeding.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to first aid 2. Reads, internalize and writes texts and questions related to first aid.	Guided discovery Discussion Observation Illustration	Describing the first aid for snake bites and nose bleeding	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
5	6	Keeping rabbits	RABBITS	RABBITS - A female rabbit is called a doe. - A male rabbit is	The learner; 1. Identifies the external parts of a rabbit.	The learner; 1. Pronounces, spells, reads writes and	Guided discovery Discussion	Describing the first aid for snake bites and	Appreciation. Care	Chart showing parts of a	Mk. Int. sci. pbk	

				called a buck - The young of a rabbit is called a kit. - Many young rabbits born at the same time are called a litter. The external parts of a rabbit. 	2. Draws and labels the external parts of a rabbit.	demonstrates meaning of words related to rabbits 2. Reads, internalize and writes texts and questions related to rabbits	Observation Illustration	nose bleeding	Awareness Fluency Concern	rabbit.	4 Int. sci. syllabus bk 4	
6	1	Keeping rabbits	Types of Rabbits	Types of Rabbits 1. local breeds 2. exotic breeds Local breeds of rabbits. ⇒ They are also called indigenous breeds or native breeds ⇒ They mainly live in bushes (burrows) on their own. Exotic breeds of rabbits. ⇒ They are kept in homes (hutches) ⇒ They live in	The learner; 1. Identifies the types of rabbits. 2. Describe the CCCs of local and exotic breeds of rabbits.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to rabbits 2. Reads, internalize and writes texts and questions related to rabbits	Guided discovery Discussion Observation Illustration	. Identifies the types of rabbits. 2. Describe the CCCs of local and exotic breeds of rabbits.	Appreciation. Care Awareness Fluency Concern	Chart showing different breeds exotic breeds.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

				hutches constructed by the farmers. Examples of exotic breeds of rabbits. 1. Angora rabbits 2. Rex rabbit 3. Californian rabbits								
6	2	Keeping rabbits	Uses of rabbits	Uses of rabbits Reasons why people keep rabbits ⇒ Rabbits are sold by farmers to get money. ⇒ Rabbits are a source of meat ⇒ Rabbits provide skins used to make bags, jackets, belts, shoes and host. Comparing the local and exotic breeds	The learner; 1. States the uses of rabbits to people. 2. Compares local breeds of rabbits and exotic breeds.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to rabbits 2. Reads, internalize and writes texts and questions related to rabbits	Guided discovery Discussion Observation Illustration	Stating the uses of rabbits to people. Comparing local breeds of rabbits and exotic breeds.	Appreciation. Care Awareness Fluency Concern	Chart showing different breeds exotic breeds.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
6	3	Keeping rabbits	Building a rabbit hutch	Building a rabbit hutch ⇒ The house of a domestic rabbit is called a hutch. ⇒ A hutch should be able to protect rabbits from easy spread of	The learner; 1. Describes the qualities of a good hutch. 2. Mentions the types of hatches. 3. draws and name the types	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to rabbits 2. Reads,	Guided discovery Discussion Observation	Describing the qualities of a good hutch. Mentioning the types of hatches. Drawing	Appreciation. Care Awareness	Chart showing different types of hutches .	Mk. Int. sci. pbk 4	

				diseases. Types of hutches. There are three types of hutches; 1. A traditional hutch 2. A morant hutch 3. A caged hutch A morant hutch 	of hatches.	internalize and writes texts and questions related to rabbits	Illustration	and name the types of hatches.	Fluency Concern		Int. sci. syllabus bk 4	
6	4	Keeping rabbits	Qualities of a good hutch	Qualities of a good hutch ⇒ It should always be clean and dry. ⇒ It should have enough air entering it. ⇒ (A wire mesh allows free circulation of air in the hutch) ⇒ It should be strong enough to protect the rabbits from wild animals and rain.	The learner; 1. Describes the qualities of a good hutch.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to rabbits 2. Reads, internalize and writes texts and questions related to rabbits	Guided discovery Discussion Observation Illustration	Describing the qualities of a good hutch. Mentioning the types of hatches. Drawing and name the types of hatches.	Appreciation. Care Awareness Fluency Concern	Chart showing different types of hutches	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

6	5		Making a rabbit hutch	Making a rabbit hutch What you need; Small sticks Banana fibres paper box glue What to do: 1. Tie small sticks in two triangular shapes. 2. Get one long stick and use it to join the two triangular shapes. 3. Make a rectangular frame from sticks. Use banana fibres to make a net as shown. 4. Place the frame in number 2 onto the rectangular frame in number 3. Join the edges using banana fibres. 6. Follow the same...	The learner; 1. Describes the steps involved when making a local caged hutch.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to rabbits 2. Reads, internalize and writes texts and questions related to housing rabbits	Guided discovery Discussion Observation Illustration	Describing the steps involved when making a local caged hutch.	Appreciation. Care Awareness Fluency Concern	Chart showing different types of hutches	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
5	6	Keeping rabbits	Managing rabbits Feeding	Managing rabbits Feeding rabbits. - Rabbits feed on different foods which	The learner; 1. Describes the steps involved when making a	The learner; 1. Pronounces, spells, reads writes and	Guided discovery Discussion	Describing the steps involved when	Appreciation. Care	Chart showing different	Mk. Int. sci. pbk	

			g rabbits .	end up providing a balanced diet to them. e.g. Pellets (these are commercial feeds) - banana peelings - sweet potato vines - leafy vegetables e.g. cabbage , carrots Breeding of rabbits. Breeding is the process through which animals give birth to their young ones. Signs of a doe on heat. -It moves up and down, all the time restless. - It lies on one side of its body. - It rubs its body against the hutch walls. NB: - A doe takes 30 days to produce. - A nest is prepared in the hutch.	local caged hutch.	demonstrates meaning of words related to housing rabbits 2. Reads, internalize and writes texts and questions related to housing rabbits	Observation Illustration	making a local caged hutch.	Awareness Fluency Concern	† types of hutches	4	
6	1	Keepin g	Hygien e in	Hygiene in rabbits ⇒ Rabbits should	The learner; 1. Describes the	The learner; 1. Pronounces,	Guided discovery	Describing the ways of	Appreciati- on.	Chalkb oard	Mk. Int.	

		rabbits	rabbits	be kept in clean places and should be given clean drinking water. ⇒ The hutch should be cleaned regularly. ⇒ Feeding container should be cleaned very often.	ways of promoting hygiene in a rabbit's hutch.	spells, reads writes and demonstrates meaning of words related to hygiene in rabbits. 2. Reads, internalize and writes texts and questions related to hygiene in rabbits.	Discussion Observation Illustration	promoting hygiene in a rabbit hutch.	Care Awareness Fluency Concern	illustration.	sci. pbk 4 Int. sci. syllabus bk 4	
6	2	Keeping rabbits	Common diseases of rabbits	Common disease of rabbits. - Ear canker - Coccidiosis - Colds - scuffles Signs and symptoms/ prevention and Treatment of common diseases of rabbits.	The learner; 1. Identifies the common diseases of rabbits. 2. States the signs and symptoms diseases of rabbits.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to diseases of rabbits. 2. Reads, internalize and writes texts and questions related to diseases of rabbits.	Guided discovery Discussion Observation Illustration	Identifying the common diseases of rabbits. Stating the signs and symptoms diseases of rabbits.	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

6	3	Keepin g rabbits	Keepin g Rabbit Record s	Keeping Rabbit Records ⇒ Records refer to written information on various activities carried out on a farm. Types of records in rabbit keeping 1. Sales and expenditure records 2. Labour records 3. Health records Other records ⇒ Number of rabbits on a farm. ⇒ Date of mating so as to prepare the nesting boxes where the doe will produce from. Importance of record keeping. ⇒ It enables a farmer to know whether he is making profits or losses. ⇒ It enables a farmer to know when to carry out	The learner; 1. Defines the term records. 2. Mentions the types of records kept on a rabbit farm. 3.. States the importance of keeping records kept on a rabbit	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to record keeping. 2. Reads, internalize and writes texts and questions related to record keeping.	Guided discovery Discussion Observation Illustration	Defining the term records. Mentioning the types of records kept on a rabbit farm. Stating the importance of keeping records kept on a rabbit	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
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				some farm activities. ⇒ It enables a farmer to get loans from banks.								
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**ARALL EDUCATIONAL
CONSULTANTS [AEC]
PRIMARY FOUR SCIENCE SCHEME OF WORK FOR TERM II, 2025**

W K	P D	THEME	S/TOPIC	CONTENT	SUBJ. COMP'NCES	LANG. COMP'NCES	METHODS	ACTIVITIES	IND. OF LSV	INST.MAT	REF.	REM
1	1	OUR FOOD	CLASSES OF FOOD	CLASSES OF FOOD -Go Foods -Grow Foods -Glow Foods Examples of each class of food. <u>Go foods:-</u> maize, cassava , millet, bread <u>Grow foods:-</u> fish, meat , chicken, beans <u>Glow foods:-</u> fruits fresh leafy vegetables	The learner; -states classes of food -gives examples of class of food	The learner; Pronounces spells, reads, writes and demonstrates meaning of words related to classes of food.	Brain storming Guided discovery Class discussion	-reading and spelling words, -drawing the examples of the given foods.	Awareness, Care, appreciation	millet seeds cassava tubers, maize flour	Mk integrated science bk 4	
1	2	Our food	food values	<u>Food values</u> These are useful food substances that the body needs to live. <u>examples:</u> carbohydrates -vitamins -proteins -fats and oils -mineral salts <u>Carbohydrates</u> -They are also called energy giving foods.(GO- foods) -They provide energy	The learner; -names the food values -defines carbohydrates -gives examples to carbohydrates -state the importance of carbohydrates to the body.	The learner; Pronounces spells, reads, writes and demonstrates meaning of words related to food values and carbohydrates.	Brain storming Guided discovery Class discussion observation	Observing some examples of food stuffs rich in carbohydrates.	- Fluency , - observation, - care, awareness	millet seeds cassava tubers, maize flour	Mk integrated science bk 4	

				to the body. <u>Examples of food stuffs that are rich in carbohydrates</u> Millet, maize, cassava, sugar, bread, sweet potato.								
1	3	Our food	vitamins and mineral salts <u>vitamins and mineral salts</u> These are healthy giving foods. <u>examples of mineral salts:</u> -calcium -phosphorous -magnesium -iodine -iron etc. <u>Food stuffs (sources of mineral: salts) fruits</u> -milk, liver, iodized salt, Vegetables, egg yolk, cereals, ripe banana, corns, etc. <u>examples of vitamins:</u> -vitamins are given names of alphabet from A to E i.e. Vitamin A,B,C,D,E <u>sources of vitamins/food stuffs with vitamins</u>	The learner; -names food stuffs that are rich in mineral salts and vitamins. -defines vitamins and mineral salts -identifies some names given to minerals salts and vitamins	The learner; Pronounces spells, reads, writes and demonstrates meaning of words related to mineral salts and vitamins heat energy to the body	Brain storming Guided discovery Class discussion observation	Observing some examples of food stuffs rich in carbohydrates.	Fluency , appreciation, care.	iodized salt fruits leafy vegetables	Mk integrated science bk 4 P.4 curriculum		

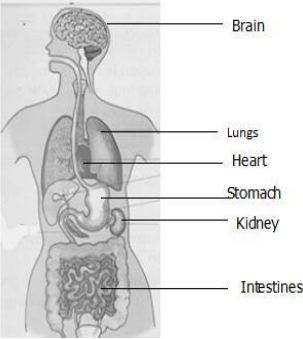
				fruits, leafy vegetables Yoghurt, carrots, etc.								
1	4	Our food	Food values	PROTEINS, FATS AND OILS. <u>proteins</u> -Proteins are body building foods. <u>uses of proteins</u> -they build body cells -they help to repair body tissues. <u>sources/ foods rich in proteins</u> fish, meat, eggs, beans, chicken <u>uses of proteins</u> -they build body cells -they help to repair body tissues. <u>sources/ foods rich in proteins</u> fish, meat, eggs, beans, chicken Fats and oils; These are energy giving foods that provide <u>sources/ foods that are rich in fats and oils</u> simsim, meat	The learner; -names food stuffs that are rich in proteins -defines proteins and fats and oils. -identifies food stuffs rich in fats and oils	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to fats and oils and proteins	Brain storming Guided discovery Class discussion observation	Observing some examples of food stuffs rich in carbohydrates	Fluency , appreciation, care	iodized salt fruits leafy vegetables	Mk integrated science bk4 4	

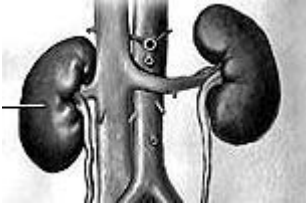
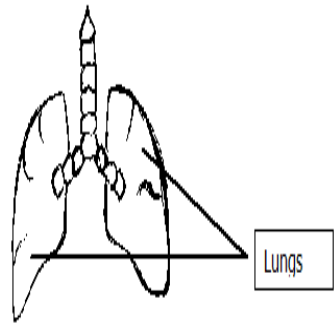
1	5	Our food	Deficiency diseases deficiency diseases; -These are diseases one gets when their bodies does not have enough of essential food values Examples of deficiency diseases -Night blindness -Kwashiorkor -Beriberi -Pellagra -Marasmus -Scurvy -Anaemia -Goitre -Rickets KWASHIORKOR -it is cause by lack of enough proteins in the body signs of a child with kwashiorkor -swollen belly -swollen moon face	The learner; -defines deficiency diseases -identifies examples of deficiency diseases -states the cause of kwashiorkor	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to deficiency diseases	Brain storming Guided discovery Class discussion observation	Observing some examples of food stuffs rich in carbohydrates.	Fluency , appreciation, care	iodized salt fruits leafy vegetables	Mk integrated science bk4	
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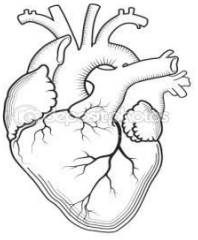
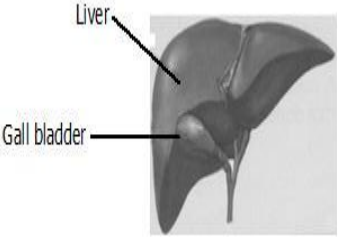
1	6	Our food	marasmus and night blindness MARASMUS -it is caused by lacked of enough carbohydrates in the body signs of a child with marasmus -swollen belly -always hungry -the face looks like that an old person -wasted muscles(thin body)  NIGHT BLINDNESS it is also called poor night vision -it is caused by shortage of vitamin A in the body prevention of night blindness -eat foods rich in vitamin A e.g. cereals	The learner; -states the cause of kwashiorkor -states the signs of marasmus in children -states the cause of night blindness -gives ways of preventing marasmus and kwashiorkor	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to night blindness, marasmus	observation, guided discovery, guided class discussion question and answer	Observing a chart showing a child suffering from marasmus	Fluency , appreciation, care	charts with children suffering from marasmus		
2	1	Our food	scurvy SCURVY -This is a deficiency disease caused by lack of vitamin C signs of scurvy -bleeding of the gums - poor healing of	The learner; -states the cause of kwashiorkor -states the signs of marasmus in children	the learner; Pronounces spells, reads, writes and demonstrates meaning of	observation, guided discovery, guided class discussion question	Observing on a chart showing different signs of beriberi	Fluency , appreciation, care	charts with children suffering from scurvy	Mk int. sci. pbk 4	

				wounds <u>BERI BERI</u> -It is caused by shortage of vitamin B ₁ in the body. <u>signs of Beriberi</u> -swelling of arms and legs -muscle spasms	-states the cause of night blindness -gives ways of preventing marasmus and kwashiorkor	words related to scurvy and beriberi	and answer				Pry. 4 sci. curr. New fount. Pbk 4	
2	3	Our food	Goitre	Goitre This is the swelling if the thyroid gland in the throat. it is caused by lack of enough of iodine mineral Rickets This a deficiency diseases caused by lack of enough of vitamin D signs of rickets in children -bones become soft and swollen -bones become and form a bow-like shape prevention of rickets -feed on feeds rick in vitamin D -expose the body on sunshine	The learner; -states the cause of kwashiorkor -states the signs of marasmus in children -states the cause of night blindness -gives ways of preventing goiter and rickets.	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to rickets and goitre.	Observation guided discovery, guided class discussion question and answer	Observing on a chart showing different signs Of rickets	Fluency , appreciation, care	charts with children suffering from rickets	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	

2	4	Our food	anemia	Anaemia tis sis a body condition caused by lack of iron or blood in the body. signs of anaemia -pale gums -white finger nails -extreme body weakness and tiredness.	The learner; -states the cause of anaemia -states the signs of anemia in children	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to anemia	observatio n, guided discovery, guided class discussion question and answer	Observing on a chart showing different signs Of anemia	Fluency , appreci ation, care	charts with children sufferin g from anemia	Mk int. sci. pbk 4 Pry. 4 sci. curr.	
2	5	Our food	Food contamination	How food gets contaminated; -serving food on dirty hands -serving food with dirty hands -leaving cooked food uncovered Dangers of food contamination -easy spread of diarrhea diseases. prevention of food contamination -washing hands before handling food -serving food in clean containers -keeping left over food well covered	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to food contamination.	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to food contamination	observatio n, guided discovery, guided class discussion question and answer	Observing on a chart showing different ways we prevent food contamin ation.	Fluency , appreci ation, care	charts showin g ways of prevent ing food contam ination	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	

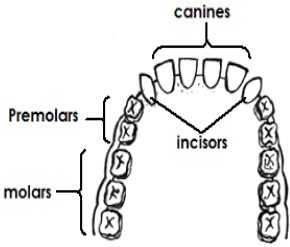
2	6	Our food	Preparation of a simple local dish at home.	How to prepare matoke locally steps: -peeling -boiling -steaming note; Matoke can also be prepared by roasting.	the learner; demonstrates ways of preparing matoke at home.	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to major organs of the human body.	observation, guided discovery, guided class discussion question and answer	-peeling and practicing preparing a local dish of matoke	self-discovery, -accuracy, self-esteem, care	Pictures showing people peeling matoke knife, banana, water,	Mk int. sci. pbk 4 Pry. 4 sci. curr. New found. Pbk 4	
3	1	Human body organs	major organs of the body	Major organs of the body -eyes, -ears, -nose, -brain, - kidney, -stomach (diagram) <u>Internal body organs</u> 	the learner;- names the major body organs -draws the structure that shows the major organs of the human body.	the learner; Pronounces spells, reads, writes and demonstrates meaning of words related to human body organs	observation, guided discovery, guided class discussion question and answer		self-discovery, -accuracy, self-esteem, care	a chart with d	Mk int. sci. pbk 4 Pry. 4 sci. curr. New found. Pbk 4	

3	2	Human body organs	kidneys	Human beings have two bean shaped kidneys. The structure of the kidneys. 	The learner; 1. Describes kidneys. 2. Draws and labels the parts of kidneys.	The learner; 1.pronounces,s pells,reads,writes and demonstrates meaning of words related to kidneys	Brain storming Guided discovery Class discussion	Drawing the structure of the kidney	self-discovery, - accuracy, self-esteem, care	A chart showing the structure of the kidneys	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	
3	3	Human body organs	<u>lungs</u>	The lungs There are two lungs in the human body The structure of lungs 	The learner; 1. Draws and labels the structure of lungs.	The learner; Pronounces,spells,reads writes and demonstrates meaning of words related to lungs.	Brain storming Guided discovery Class discussion	Drawing the structure of the lungs	Critical thinking Appreciation Fluency Care	An illustration of the lungs on a chart.	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	

3	4	Human body organs	The heart	The lungs The heart is located in the chest cavity between the lungs. <u>Structure of the heart</u> 	The learner; 1.Draws the structure of the heart. 2. Answers the given questions on the heart.	The learner; Pronounces,spells reads, reads writes and demonstrates meaning of words related to the heart.	Observation. Guided discovery Class discussion	Drawing the structure of the heart	Fluency Appreciation Care Kindness	Drawing diagrams showing lungs	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	
3	5	Human body organs	The liver	The liver - This is found below the lungs in the abdominal cavity - The liver produces bile which helps to break down food into small particles. - 	The learner; 1. Draws the structure of the liver. 2. Answers given questions on the liver	The learner;	Observation. Guided discovery Class discussion	Observation Drawing the structure of the liver.	Fluency Appreciation Care Kindness		Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	

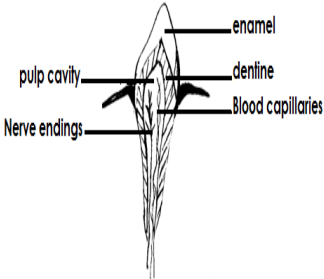
3	6	Human body organs	The stomach	The structure of the stomach - The stomach is one of the organs of the digestive system - The stomach stores food for some time Structure of the stomach  stomach	The learner; 1.draws the structure of the stomach 2.answers the questions	The learner; Pronounces,spells,read,writes and demonstrates words related to the liver.	Observation. Guided discovery Class discussion	Drawing the structure of the stomach.	Drawing diagrams showing lungs.	a chart with diagrams of the stomach	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	
4	1	Human body organs	The brain of the brain	This is the soft organ protected by the skull. The brain controls all activities of the body Structure Structure of the brain  Brain <u>Functions of the brain</u> It stores information For thinking For reasoning	The learner; 1. Draws the structure of the brain 2. Gives the functions of the brain.	The learner; Pronounces,spells,read,writes and demonstrates words related to the brain	Observation. Guided discovery Class discussion	Drawing the structure of the brain	Critical thinking Appreciation fluency	a chart with diagrams of the brain	Mk int. sci. pbk 4 Pry. 4 sci. curr. New fount. Pbk 4	

4	2	Human body organs	How the Human body Works	How the human body works ⇒ The human body takes in food through the mouth oxygen during breathing. ⇒ Breathing is the taking in and out of oxygen. ⇒ The human body transports food and oxygen to different parts of the body. ⇒ The human body uses food and oxygen to get energy. Food and oxygen is moved around the body by the blood. ⇒ The human body removes wastes Examples of waste materials ⇒ Urine ⇒ sweat ⇒ carbon dioxide	The learner; 1. describes how the body works. 2. Mentions the examples of waste materials.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to how the human body works. 2. Reads, internalize and writes texts and questions to related how the human body works.	Guided discovery Discussion observation	Describing how the human body works. Mentioning example of waste material of the human body.	Appreciation. Care Awareness Fluency Concern	Chalkboard illustration.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
4	3	The teeth	Sets of teeth	Sets of teeth ⇒ Humans beings grow two sets of teeth in their life time namely, ⇒ Milk teeth	The learner, 1. Describes the sets of teeth. 2. Describes the milk teeth	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of	Guided discovery Discussion	Describing the sets of teeth i.e. The milk teeth and	Appreciation. Care	A chart showing sets of teeth	Mk. Int. sci. pbk 4	

				⇒ Permanent teeth Milk teeth ⇒ Milk teeth are the first set of teeth in children. ⇒ They are twenty in number ⇒ They start appearing in children at the age of 5-9 months ⇒ Milk teeth start falling out at the age of 5-6 years ⇒ The incisors are the first to grow and the first to fall out 	and the permanent teeth	words related sets of teeth e.g. Milk teeth, permanent, canine, incisors and premolars. 2. Reads, internalize and writes texts and questions related to sets of teeth.	observation	permanent teeth.	Awareness Fluency Concern		Int. sci. syllabus bk 4	
4	4	The teeth	Types of teeth	Types of teeth i) Canines ii) Incisors iii) Pre Molar iv) Molars Canines ⇒ Canines are sharp, and pointed. ⇒ They are used for tearing food.	The learner; 1. States the types of teeth. 2. Describes the canine and incisor teeth. 3. States the importance of the canine and incisor teeth.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related types of teeth e.g. canine,	Guided discovery Discussion Observation	Stating the types of teeth. Describing the incisors and canine	Appreciation. Care Awareness	A chart showing incisors and canine teeth.	Mk. Int. sci. pbk 4 Int.	

				Structure of a canine tooth  ⇒ <u>Adaptation to its function</u> ⇒ They are sharp and pointed for tearing food Incisors ⇒ They are chisel shaped ⇒ They have one root Structure of an incisor  <u>Adaptation to its function</u> ⇒ They are chisel shaped for cutting. ⇒ There are four incisors in each jaw.	4. Describes the adaptation of the canines and incisor teeth to their function.	incisors, molars and premolars. 2. Reads, internalize and writes texts and questions related to sets of teeth.	n	teeth. Stating the functions of the canine and incisor teeth. Describing the adaptations of the canines and incisor teeth to their functions.	Fluency Concern		sci. syllabus bk 4	
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4	5	Types of teeth	Pre molars And Molars Pre molars ⇒ They are broad at the top with two cups/ridges ⇒ They have one root. Structure of a pre molar  <u>Adaptation to its function</u> ⇒ They have broad tops with ridges for crushing food. • Molars ⇒ They have broad wide tops with 3-5 ridges. Structure of a molar <u>Adaptation to its function</u> ⇒ They have broad tops with cups for crushing food	The learner; 2. Describes the premolars and molar teeth. 3. States the importance of the pre molars and molars teeth. 4. Describes the adaptation of the molars and pre molars to their function.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related types of teeth e.g. canine, incisors, molars and premolars. 2. Reads, internalize and writes texts and questions related to sets of teeth.	Guided discovery Discussion Observation	Describing the molars and pre molars teeth. Stating the functions of the molars and pre molars teeth. Describing the adaptations of the molars and pre molars teeth. to their functions.	Appreciation. Care Awareness Fluency Concern	A chart showing molars and pre molars teeth.	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
4	6	The teeth	Regions of the teeth Regions of a tooth ⇒ A tooth has three regions namely; a) The crown b) The neck c) The root	The learner; 2. Identifies the regions of the tooth. 3. Draws and labels the regions of the	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words	Guided discovery Discussion	Identifying regions of the tooth. Drawing	Appreciation. Care	A chart showing regions of the tooth	Mk. Int. sci. pbk 4	

					tooth.	related regions of the tooth 2. Reads, internalize and writes texts and questions related to regions of the tooth	Observation	and labeling the regions of the tooth	Awareness Fluency Concern		Int. sci. syllabus bk 4	
5	1	The teeth	Parts of a tooth	Internal structure of a tooth  The enamel The dentine The pulp cavity Nerve endings Blood capillaries	The learner; 1. Draws and labels parts of a canine and molar tooth	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related parts of the tooth 2. Reads, internalize and writes texts and questions related to parts of the tooth	Guided discovery Discussion Observation	Drawing the structure of the tooth. Labeling the parts of a tooth.	Appreciation. Care Awareness Fluency Concern	A chart showing the structure of a tooth	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	

5	2	The teeth	Parts of the tooth Functions of parts of at tooth <u>The enamel</u> ⇒ It is the hardest outer part of the tooth. ⇒ The enamel protects the inside part of the tooth to wear and tear. <u>The dentine</u> ⇒ It is the biggest part of a tooth. ⇒ The dentine contains the pulp cavity and living cells through which it receives food. <u>The pulp cavity</u> ⇒ It is the sensitive part of the tooth ⇒ The pulp cavity contains blood vessels and nerves which are sensitive to heat coldness and pain <u>Nerve endings</u> ⇒ They pick nerve signals/ messages and send them to the brain for interpretation. Blood capillaries ⇒ They send blood,	The learner' 1. States the functions of parts of the tooth.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related parts of the tooth 2. Reads, internalize and writes texts and questions related to parts of the tooth	Guided discovery Discussion Observation	Stating the functions of parts of the tooth Labeling the parts of a tooth.	Appreciation. Care Awareness Fluency Concern	A chart showing the structure of a tooth	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
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				oxygen and digested food to the tooth.								
5	3	The teeth		Care of our teeth ⇒ We need to care for our teeth and keep them healthy. ⇒ The general cleanliness for our mouth is called oral hygiene. Ways of cleaning for the teeth ⇒ Brushing the teeth every after a meal to remove the food remains. ⇒ By dental flossing to remove food particles between the teeth. Things we use to care for our teeth ⇒ Tooth brush ⇒ Tooth paste ⇒ Dental floss ⇒ Clean water	The learner; 1. Identifies the ways of caring for the teeth. 2. States the materials used to clean teeth. 3. Draws the materials used to clean teeth.	The learner; 1. Pronounces, spells, reads writes and demonstrate s meaning of words related to care for the teeth. 2. Reads, internalize and writes texts and questions related to care for the teeth.	Guided discovery Discussion Observation Illustration Demonstration	Identifyin g the ways of caring for the teeth. Stating the materials used to clean teeth. Drawing the materials used to clean teeth.	Appreciation. Care Awareness Fluency Concer n	A tooth brush, tooth paste, dental floss.	Mk. Int. sci. pbk 4 Int. sci. syllab us bk 4	

5	4	The teeth	Diseases of the teeth 1. Tooth decay(dental caries) 2. Gum diseases(gingivitis) 3. Dental plaque 4. Periodontal disease Refer to the table of diseases of the teeth in the lesson notes. Disorders of the teeth • Disorders of the teeth i) Poor growth of teeth (malocclusion) ii) broken or cracked teeth • Control of teeth disorders ⇒ Avoid biting hard things. ⇒ Avoid rough games.	The learner; 1.Outlines the diseases of the teeth. 2. States the signs and symptoms of the diseases of the teeth. 3. Mentions the ways of preventing the diseases of the teeth. 4. Identifies the disorders of the teeth and how they can be controlled.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to diseases of the teeth. 2. Reads, internalize and writes texts and questions related to diseases of the teeth.	Guided discovery Discussion Observation Illustration Demonstration	Outlining the diseases of the teeth. Stating the signs and symptoms of diseases of the teeth. Mentioning ways of preventing diseases of the teeth	Appreciation. Care Awareness Fluency Concern	Pictures of decayed teeth	Mk. Int. sci. pbk 4 Int. sci. syllabus bk 4	
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				Protozoa Fungi <u>Places where germs can be found(habitats for germs)</u> In un boiled water In the soil In air In human and animal faeces On plants In human body and other animals In dirtyplaces like latrine			Demonstration					
6	1			Ways through which germs are spread. Through drinking contaminated water. Through eating contaminated food. Through bites of insects and animals. Through breathing	The learner; 1.Mentions the ways through which germs are spread 2.names the diseases spread through the given ways on the table.	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to sanitation. 2. Reads, internalize and writes texts and questions related to sanitation.	Discussion Observation Illustration Demonstration	Mentioning ways through which germs are spread	Audibility Critical thinking fluency	A chart showing the stages in the 4fs.	Comprehensive science book 4 Primary curriculum book4 .	

				contaminated air. Through contact with people having germs. <u>Table showing ways germs are spread and their causative agents</u>							
6	2	sanitation	germs	The 4fs germ path and economic importance Is the way germs are spread by houseflies through faeces, flies, food fingers (4fs) Flies carry germs on their legs, wings and hairy bodies from faeces onto our food.	The learner; 1. writes 4fs in full. 2. describes the stages of the 4fs path.	The learner; 1. Pronounces, spells, reads, writes and demonstrates meaning of words related to 4fs. 2. Reads, internalizes and writes texts and questions related to 4fs.	Discussion Observation Illustration Demonstration	Describing stages in the 4fs germ path.	audibility Critical thinking fluency	Chart showing the 4fs germ path	Comprehensive science book 4 Primary curriculum book 4

			diseases Diseases A disease is an illness of the body which brings disorders Diseases spread due to 4fs. Typhoid Diarrhea Bilharzia. Dysentery malaria	The learner; 1. defines the term disease. 2. lists diseases that are spread through 4fs	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to diseases spread through 4fs. 2. Reads, internalize and writes texts and questions related to 4fs.	Discussion Observation Illustration Demonstration	Listing diseases that are spread through 4fs.	audibility Critical thinking fluency	Work cards showing diseases spread through 4fs.	Comprehensive science book 4 Primary curriculum book4	w
6	3		Ways of protecting against germs and diseases Keeping our homes and school clean. Cleaning and covering latrines. Keeping our kitchen clean. Covering our food properly.	The learner; 1. mentions Ways of protecting against germs and diseases	The learner; 1. Pronounces, spells, reads writes and demonstrates meaning of words related to protection against germs and diseases 2. Reads, internalize and writes texts and questions related to germs and	Discussion Observation Illustration Demonstration	.mentioning the Ways of protecting against germs and diseases	audibility Critical thinking fluency	environment	Comprehensive science book 4 Primary curriculum book4	

				Washing hands before eating food. Washing and ironing clothes and beddings. Cutting tall grass around our homes.		diseases						
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